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(54) **OPTICAL PORT/CONNECTOR MAGNETIC SECURING/RELEASE SYSTEM**

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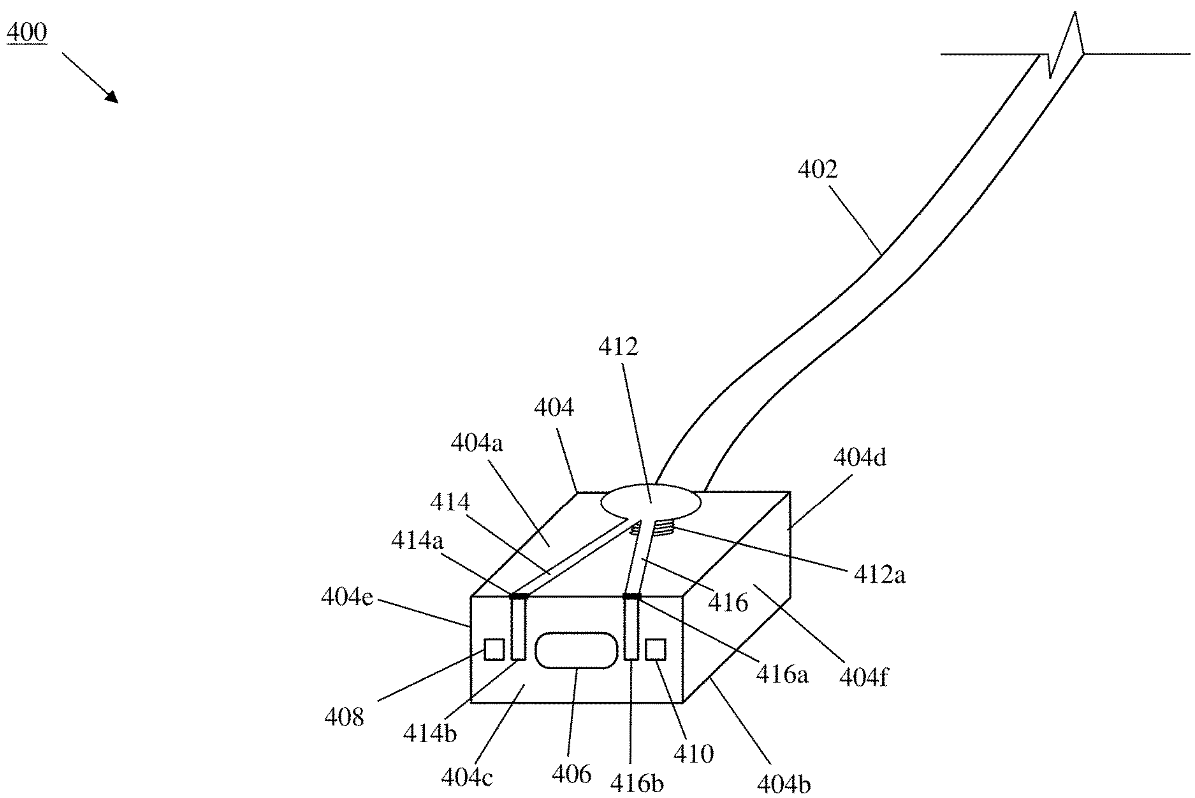
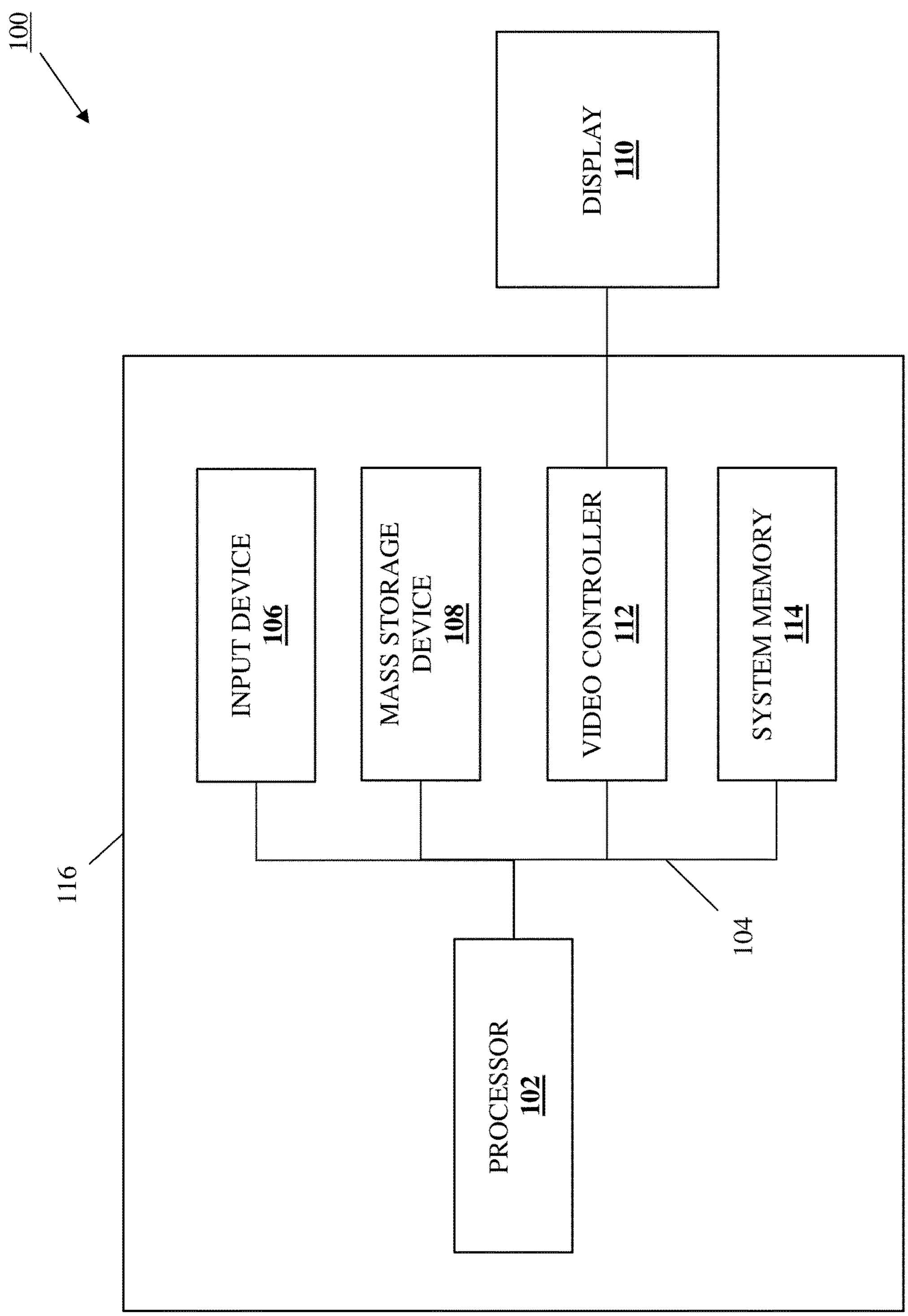
(57) **ABSTRACT**

An optical cable system includes an optical cable and an optical connector that is included on the optical cable. A connector magnetic securing subsystem is included on the optical connector and is configured to magnetically connect to a port magnetic securing subsystem on an optical port to secure the optical connector to an optical port. A mechanical release device is coupled to the optical connector and is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port.

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INPUT DEVICE 106
MASS STORAGE DEVICE 108
VIDEO CONTROLLER 112
SYSTEM MEMORY 114
116
104
PROCESSOR 102

FIG. 1

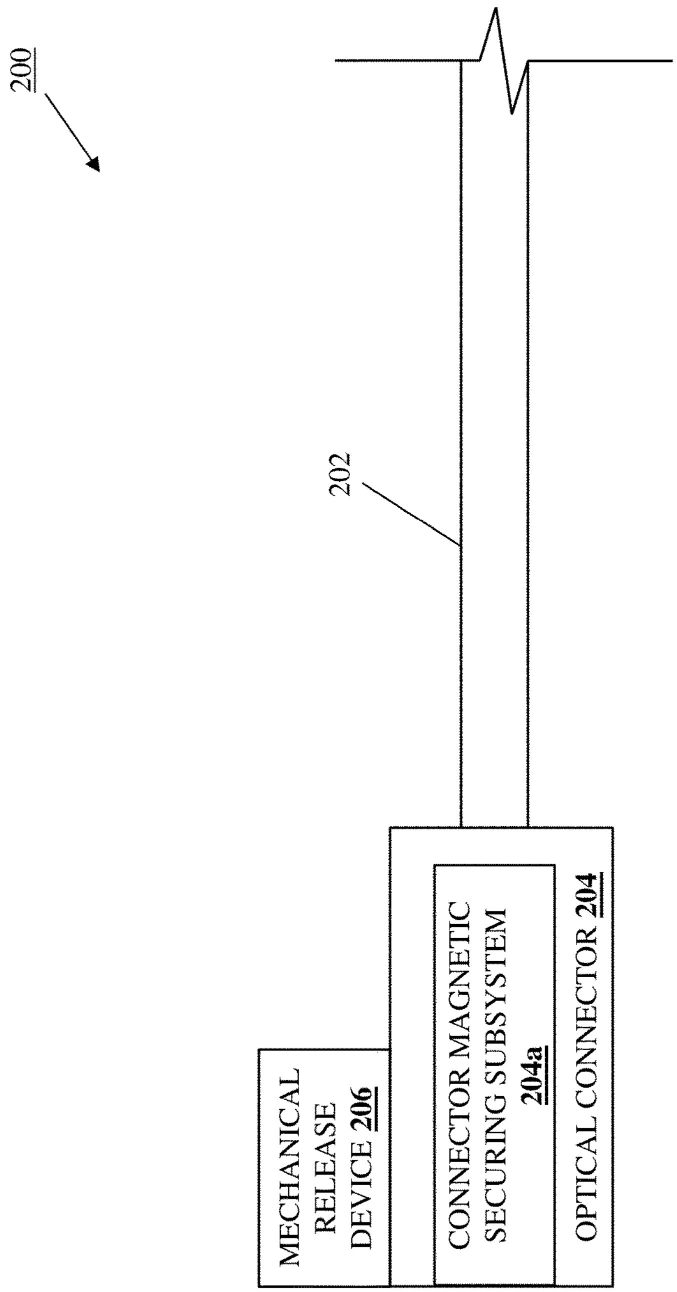

200
202
MECHANICAL RELEASE DEVICE 206
CONNECTOR MAGNETIC SECURING SUBSYSTEM 204a
OPTICAL CONNECTOR 204

FIG. 2

FIG. 3A

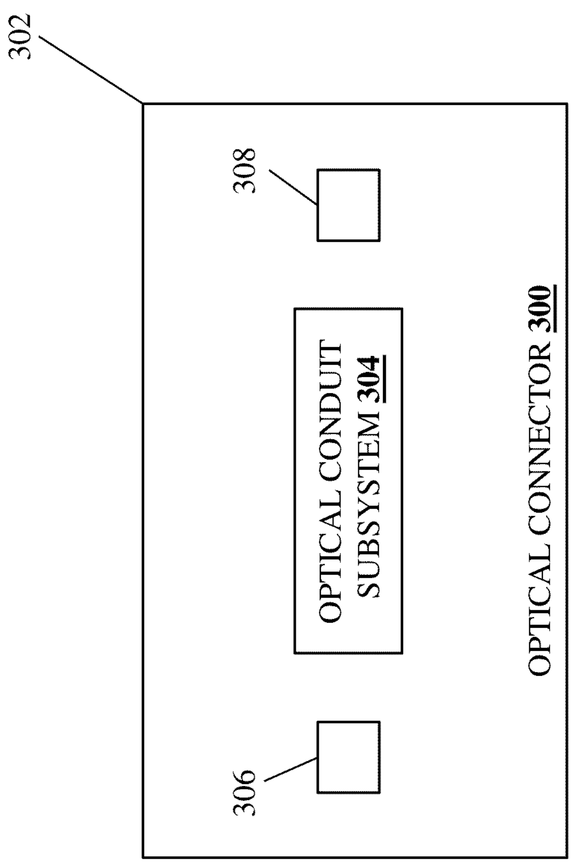
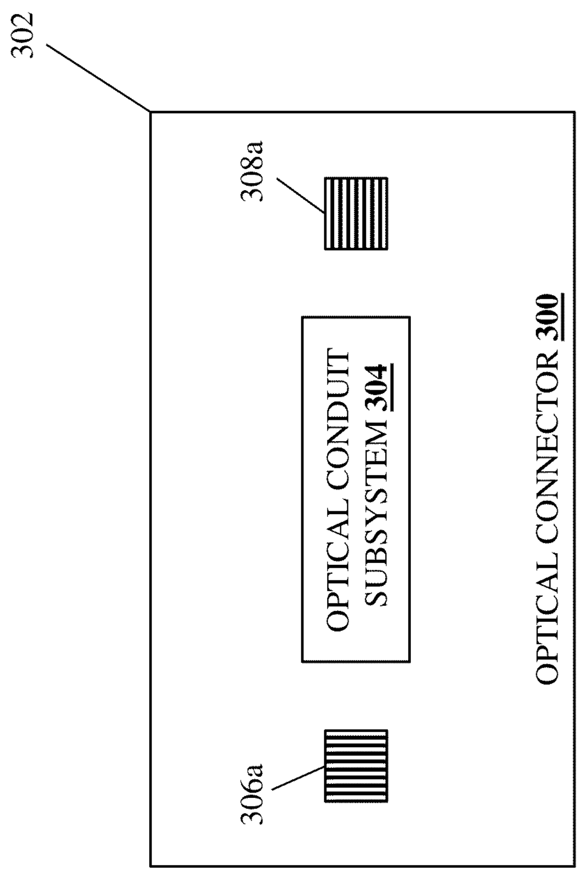

302
308
OPTICAL CONDUIT SUBSYSTEM 304
OPTICAL CONNECTOR 300
306

FIG. 3B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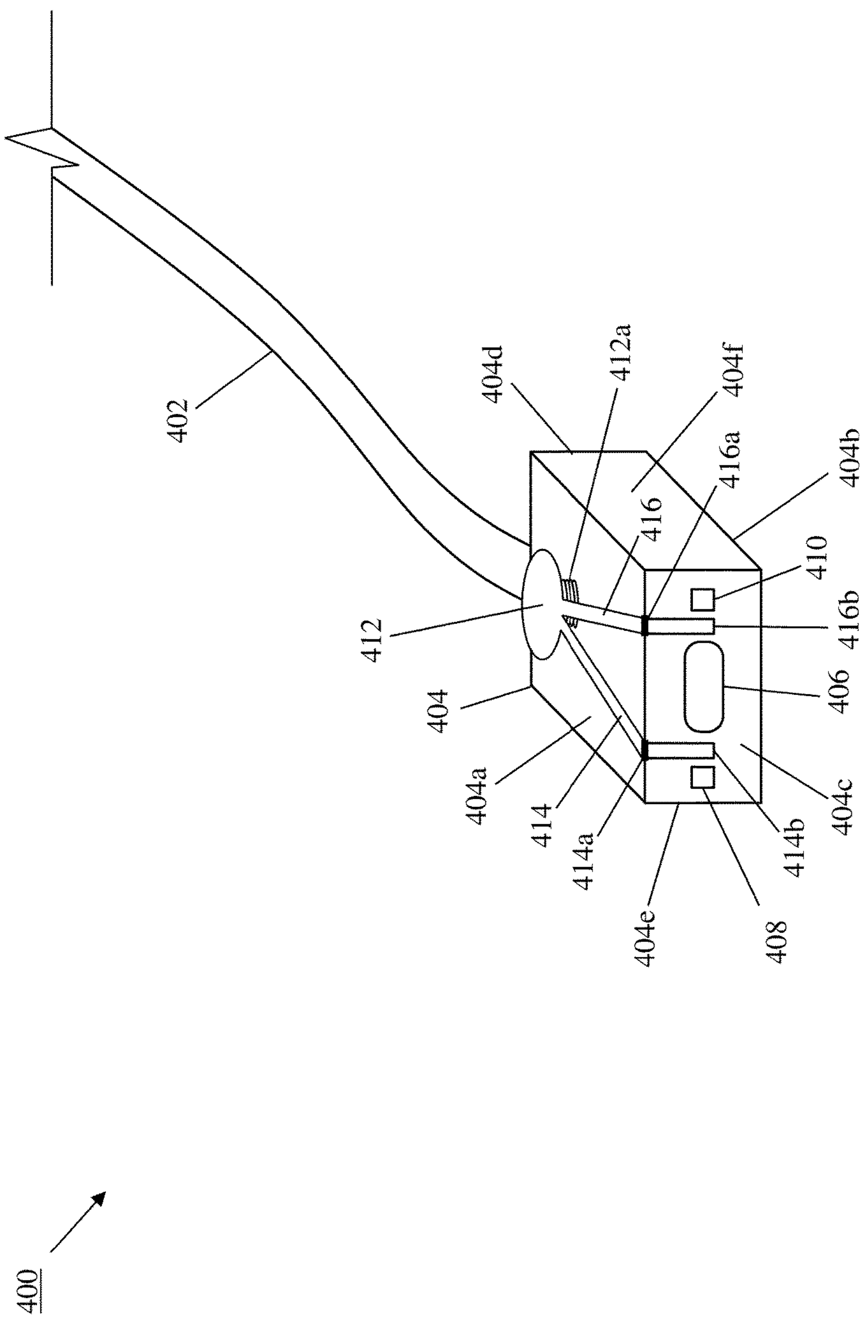
306a
OPTICAL CONNECTOR 300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404
404a
404b
404c
404d
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404f
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408
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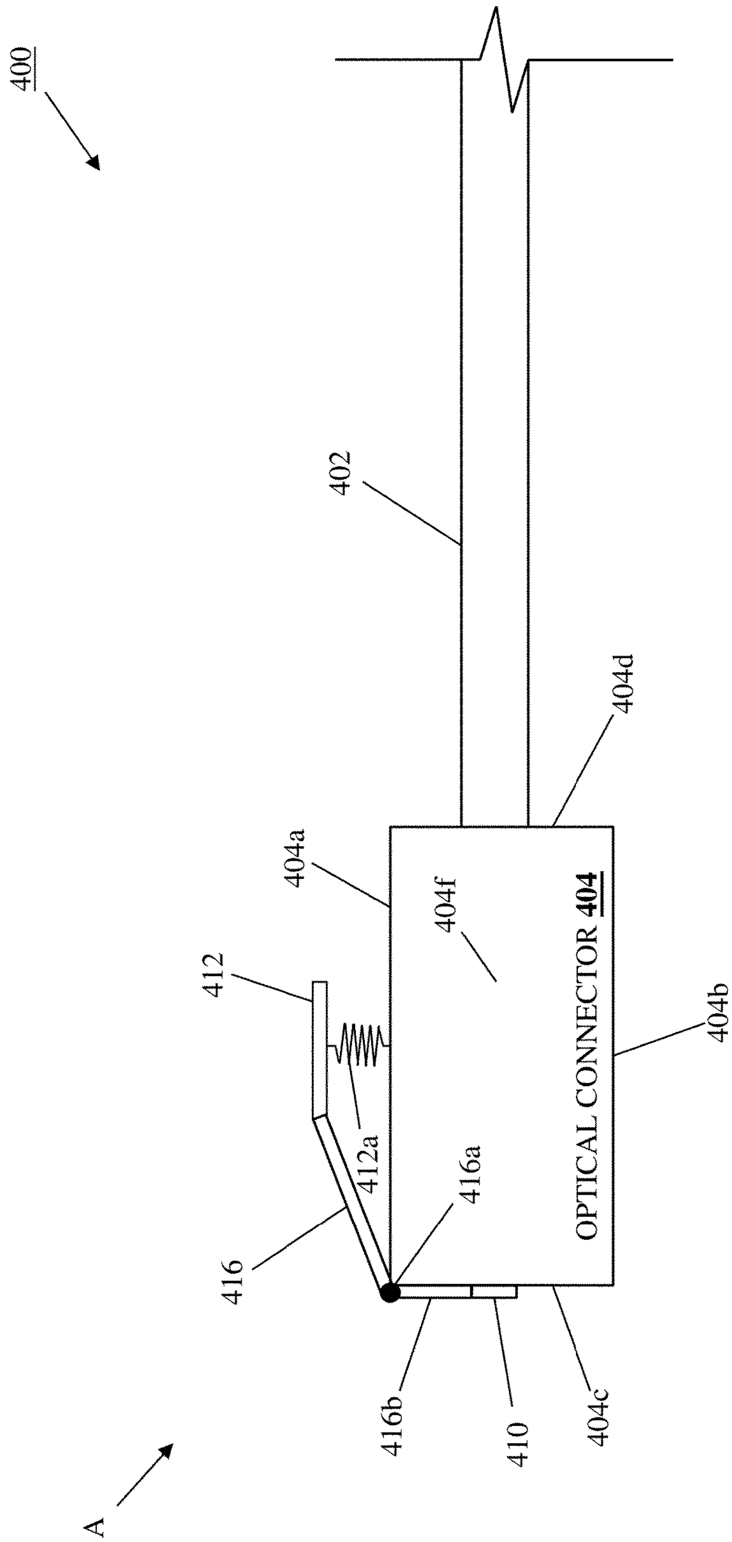
412
412a
414
414a
414b
416
416a
416b

FIG. 4A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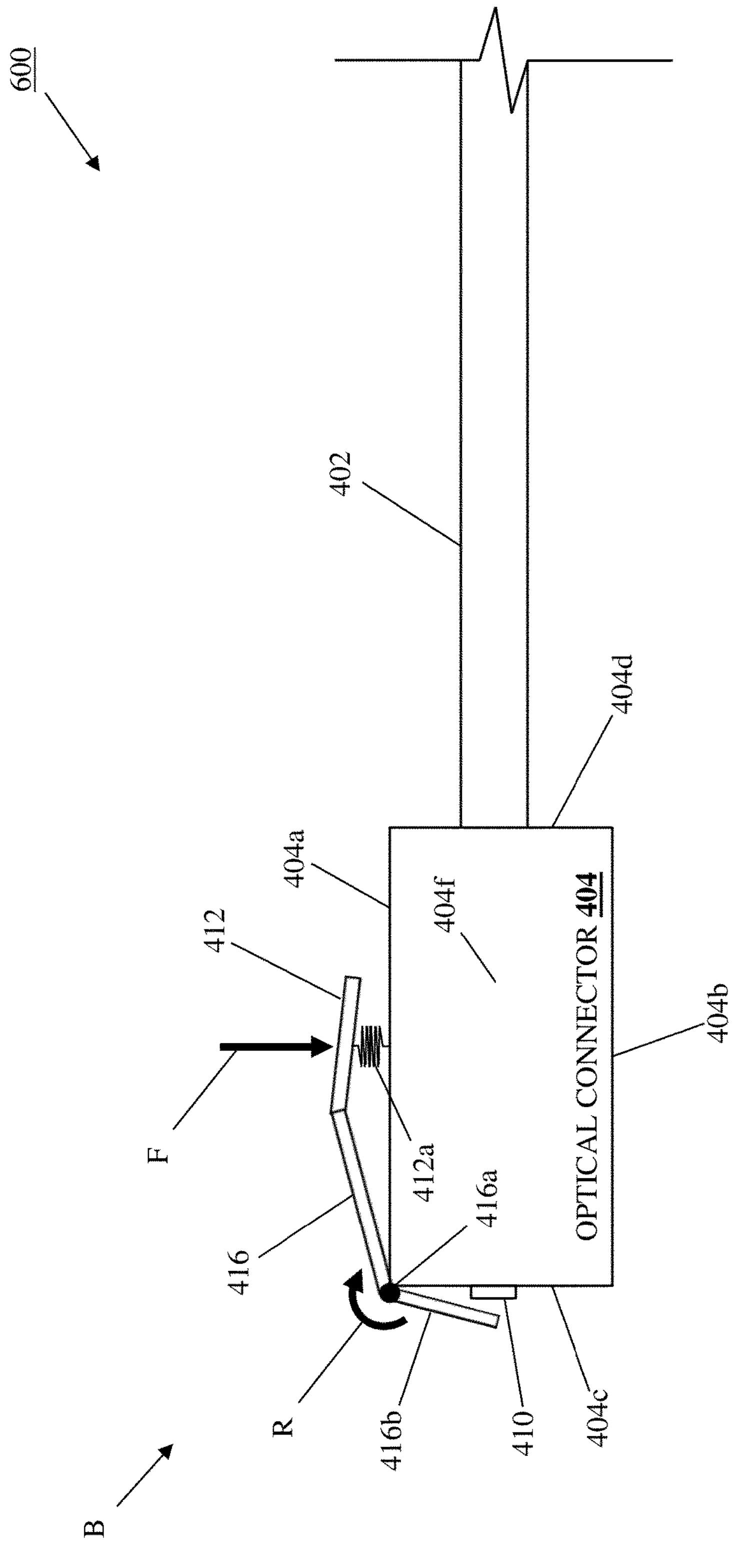
404b
412
412a
416a
416
416b
410
404c
A

FIG. 4B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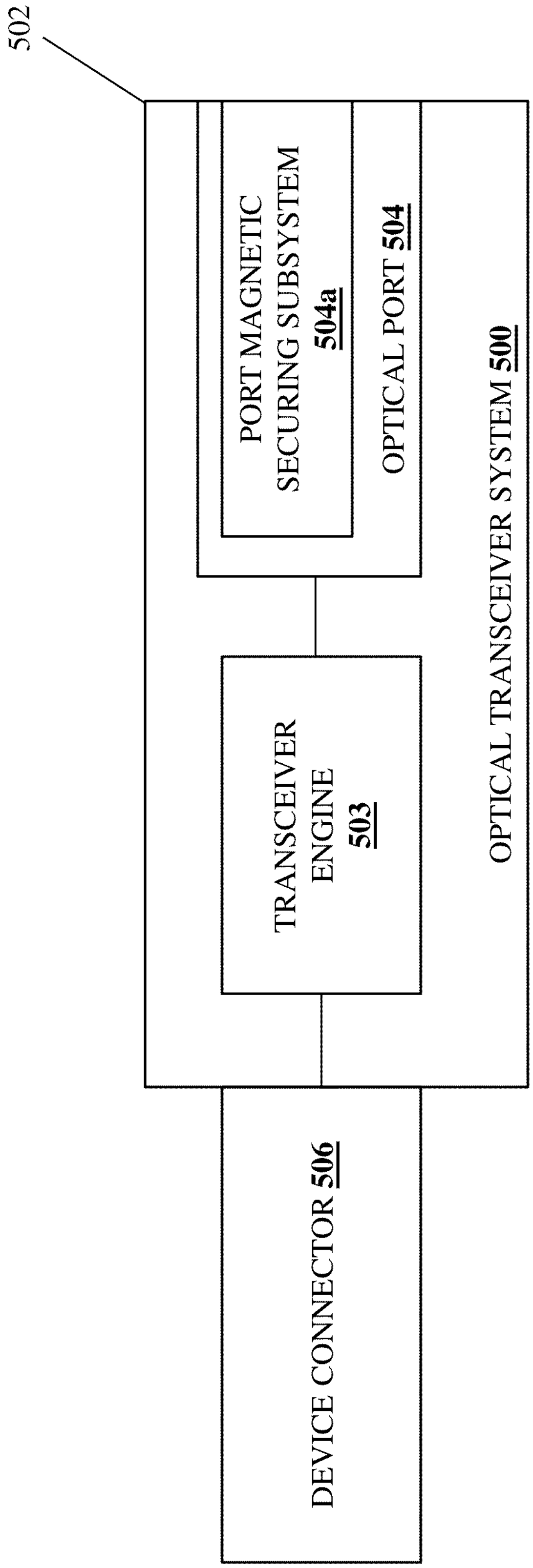
OPTICAL CONNECTOR 404
404b
F
412a
416a
416
R
416b
410
404c
B

FIG. 4C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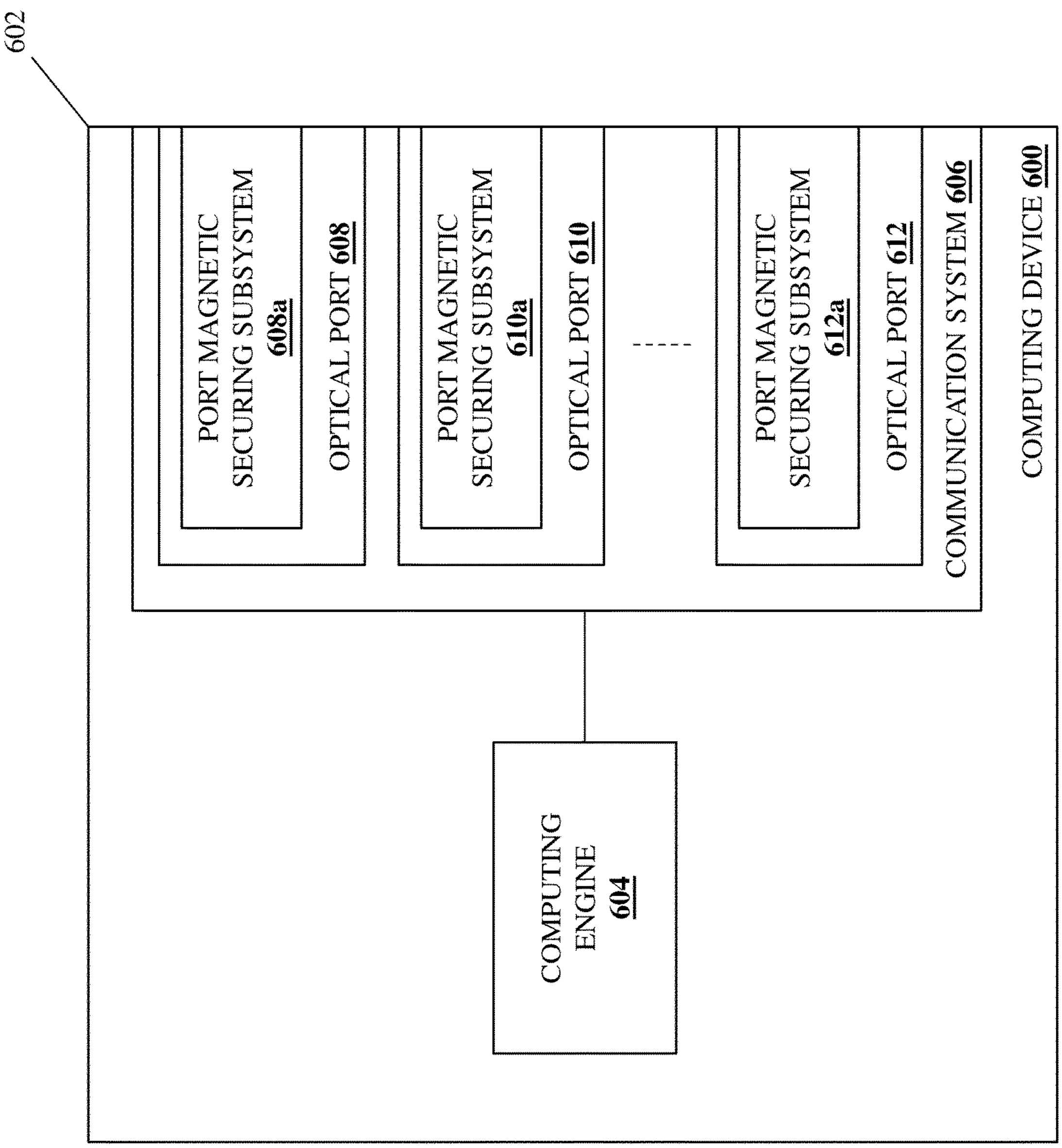
TRANSCEIVER ENGINE 503
OPTICAL TRANSCEIVER SYSTEM 500
DEVICE CONNECTOR 506

FIG. 5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PORT MAGNETIC SECURING SUBSYSTEM 610a
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PORT MAGNETIC SECURING SUBSYSTEM 612a
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 6

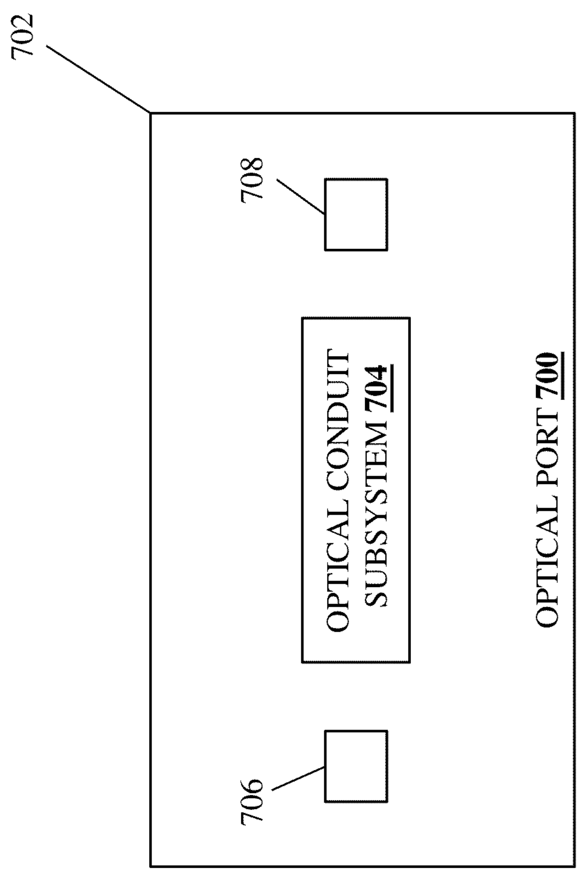


FIG. 7A

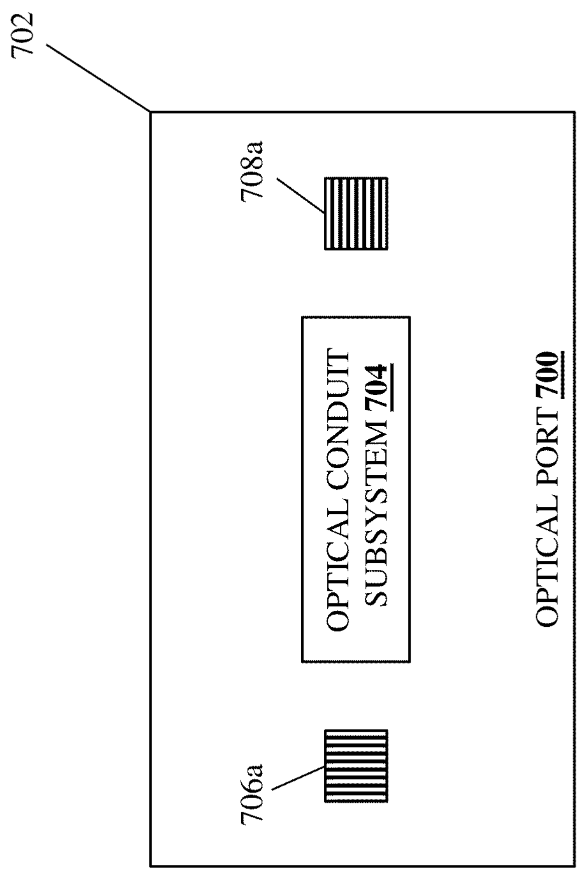

702
708a
OPTICAL CONDUIT SUBSYSTEM 704
OPTICAL PORT 700
706a

FIG. 7B

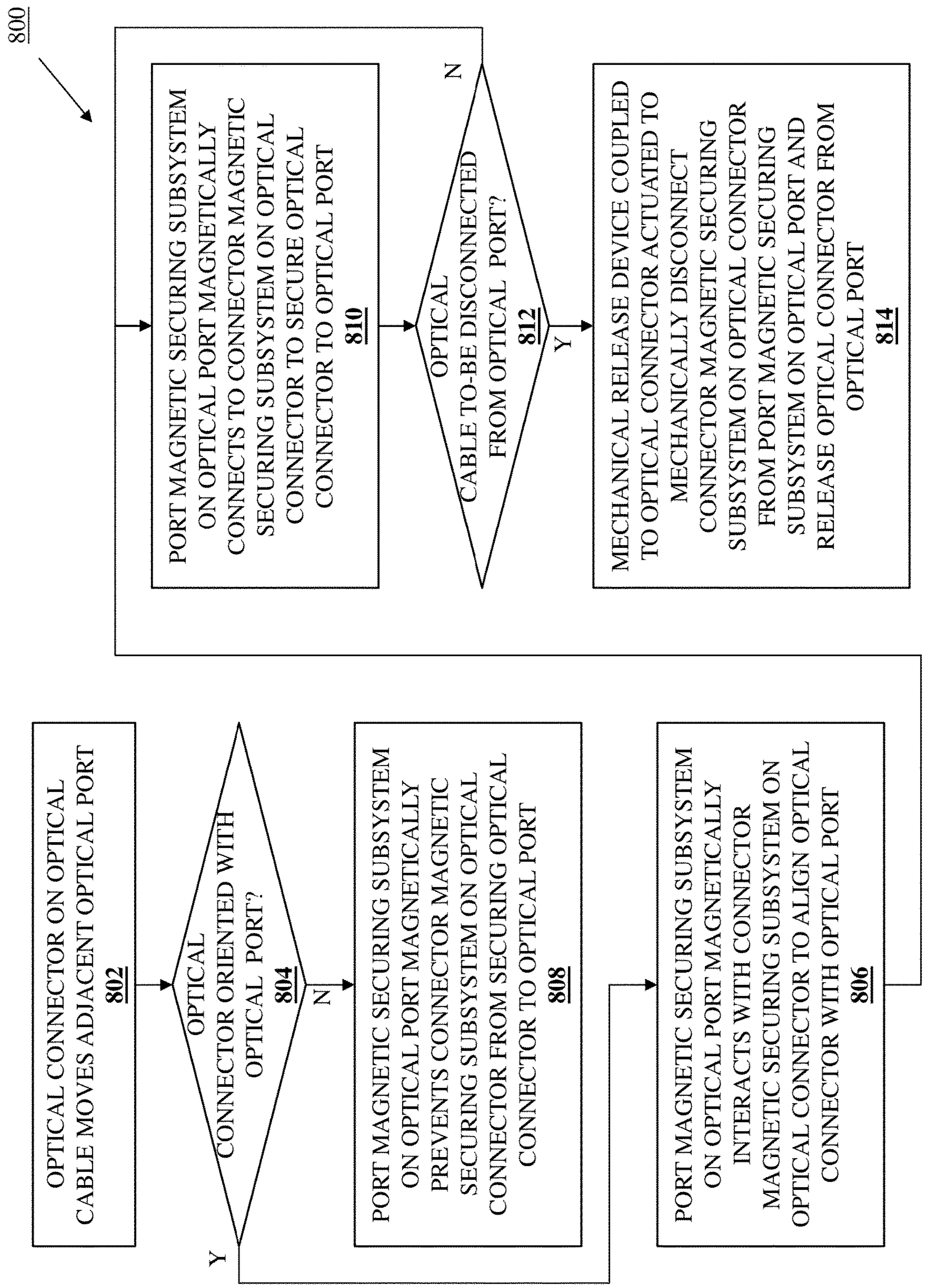
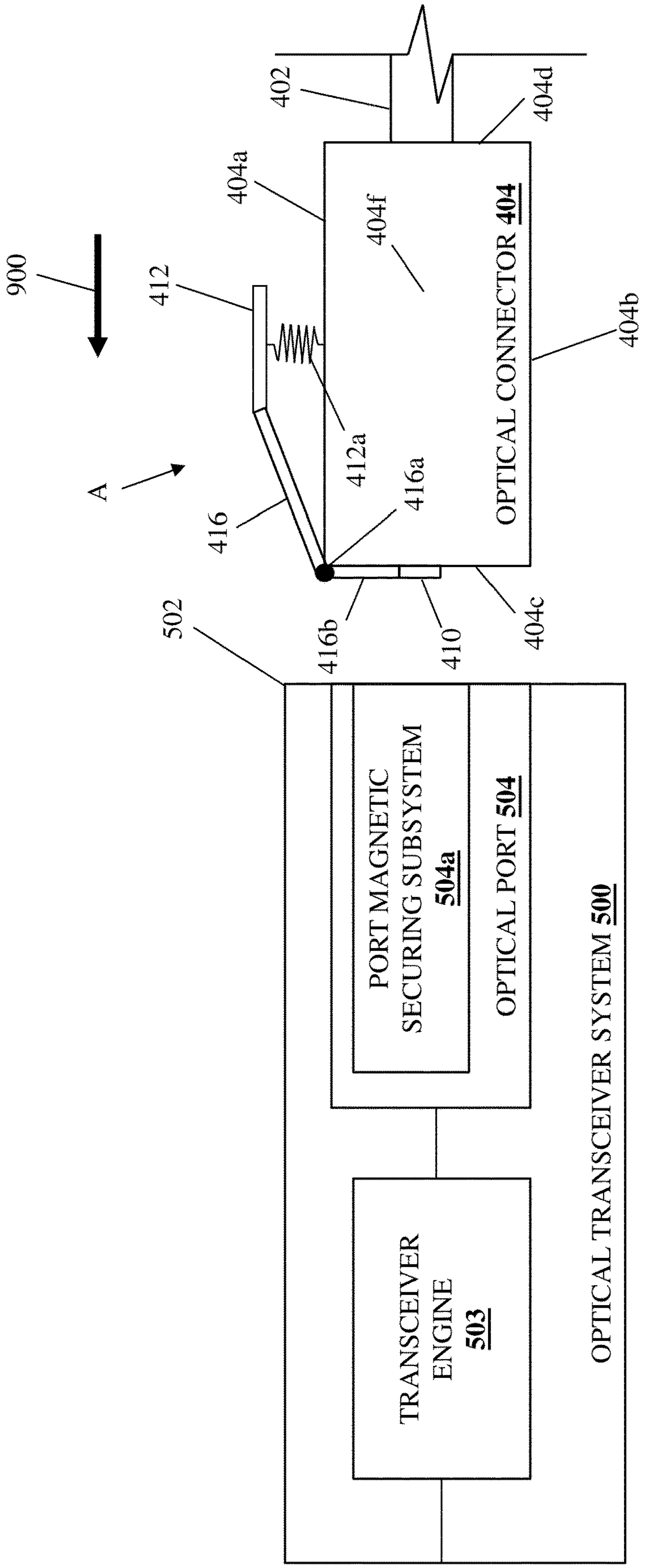

800
OPTICAL CONNECTOR ON OPTICAL CABLE MOVES ADJACENT OPTICAL PORT
802
OPTICAL CONNECTOR ORIENTED WITH OPTICAL PORT?
804
Y
N
PORT MAGNETIC SECURING SUBSYSTEM ON OPTICAL PORT MAGNETICALLY PREVENTS CONNECTOR MAGNETIC SECURING SUBSYSTEM ON OPTICAL CONNECTOR FROM SECURING OPTICAL CONNECTOR TO OPTICAL PORT
808
PORT MAGNETIC SECURING SUBSYSTEM ON OPTICAL PORT MAGNETICALLY INTERACTS WITH CONNECTOR MAGNETIC SECURING SUBSYSTEM ON OPTICAL CONNECTOR TO ALIGN OPTICAL CONNECTOR WITH OPTICAL PORT
806
PORT MAGNETIC SECURING SUBSYSTEM ON OPTICAL PORT MAGNETICALLY CONNECTS TO CONNECTOR MAGNETIC SECURING SUBSYSTEM ON OPTICAL CONNECTOR TO SECURE OPTICAL CONNECTOR TO OPTICAL PORT
810
OPTICAL CABLE TO-BE DISCONNECTED FROM OPTICAL PORT?
812
N
Y
MECHANICAL RELEASE DEVICE COUPLED TO OPTICAL CONNECTOR ACTUATED TO MECHANICALLY DISCONNECT CONNECTOR MAGNETIC SECURING SUBSYSTEM ON OPTICAL CONNECTOR FROM PORT MAGNETIC SECURING SUBSYSTEM ON OPTICAL PORT AND RELEASE OPTICAL CONNECTOR FROM OPTICAL PORT
814

FIG. 8


900
A
402
404a
404d
404f
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412a
416
416a
416b
410
404c
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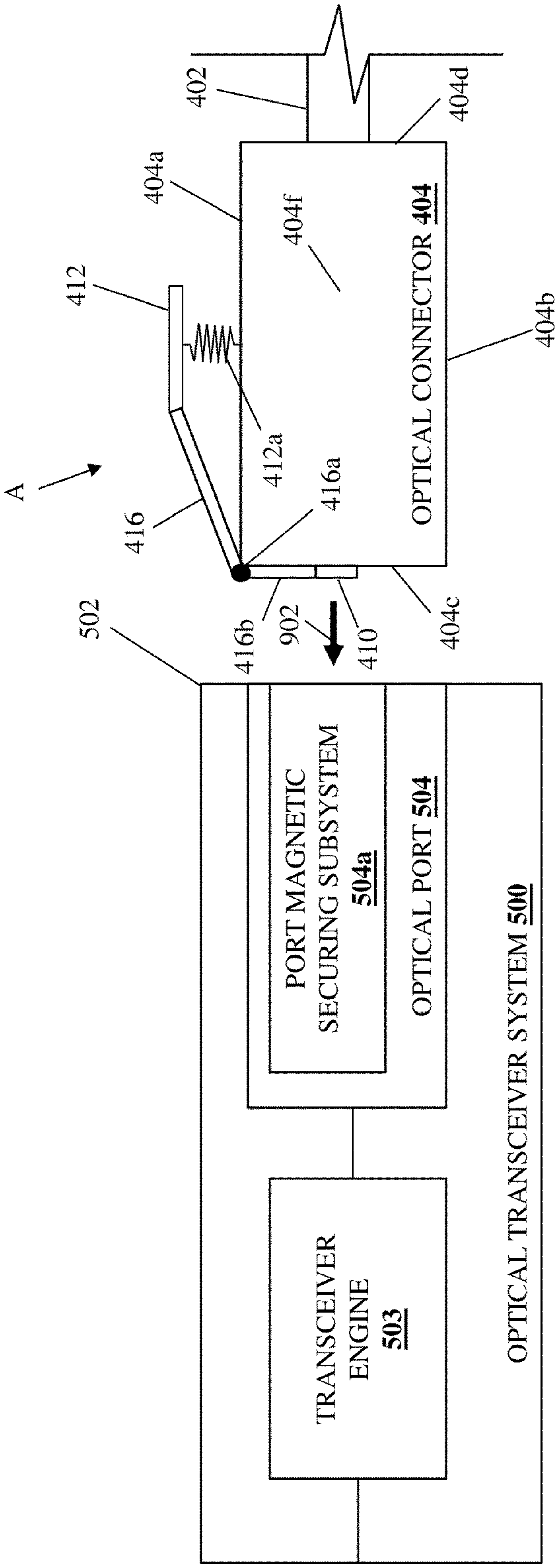
OPTICAL CONNECTOR 404
502
PORT MAGNETIC SECURING SUBSYSTEM 504a
OPTICAL PORT 504
TRANSCEIVER ENGINE 503
OPTICAL TRANSCEIVER SYSTEM 500

FIG. 9A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416b
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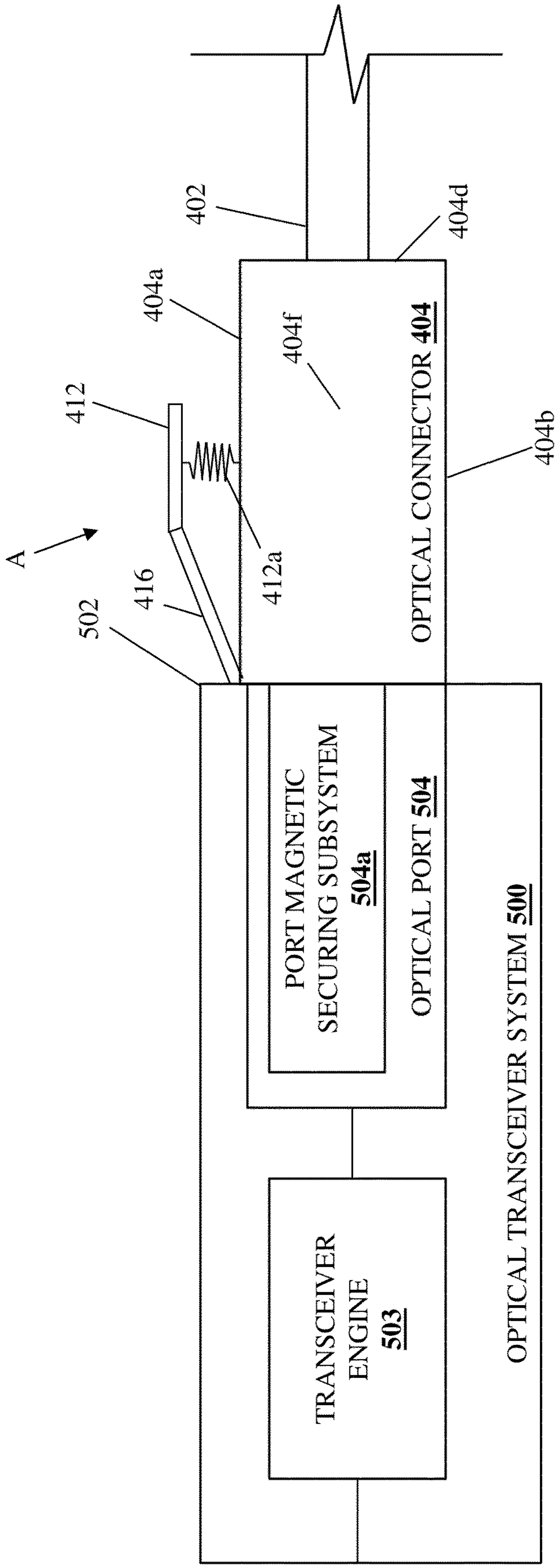
504a
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER
ENGINE
503

FIG. 9B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412a
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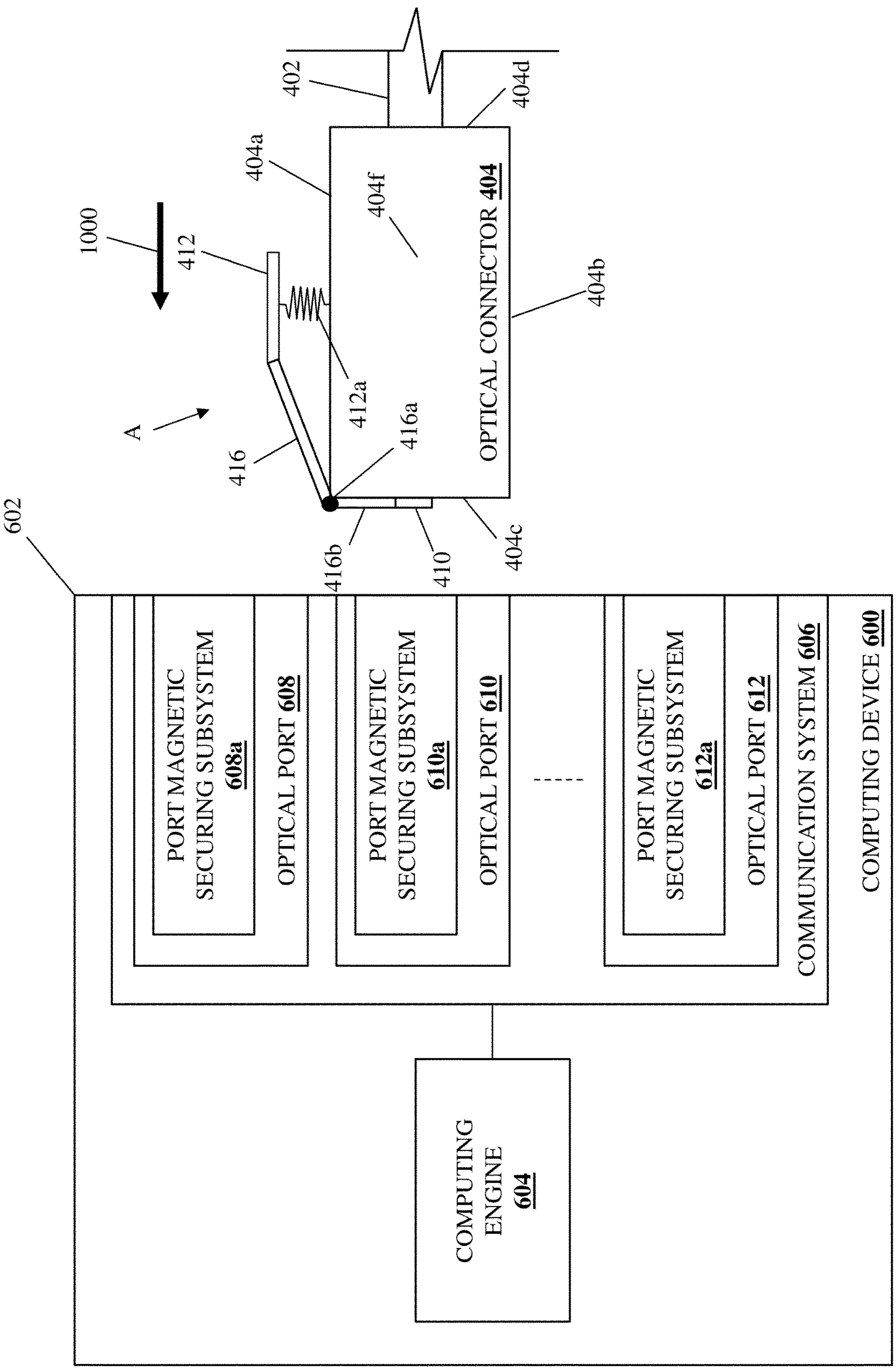
OPTICAL PORT 504
TRANSCEIVER ENGINE 503
OPTICAL TRANSCEIVER SYSTEM 500

FIG. 9C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OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
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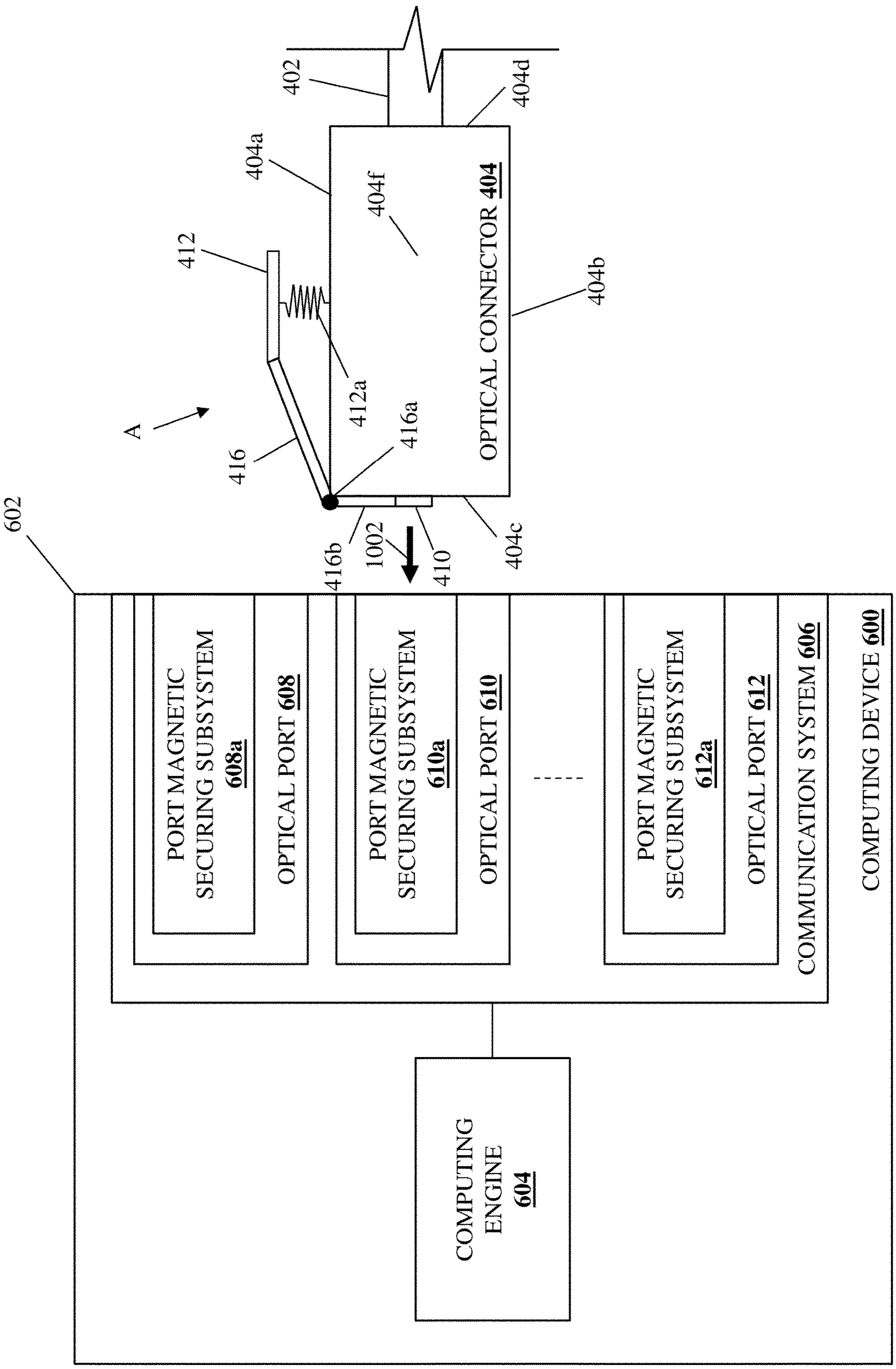
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 10A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PORT MAGNETIC SECURING SUBSYSTEM 608a
OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
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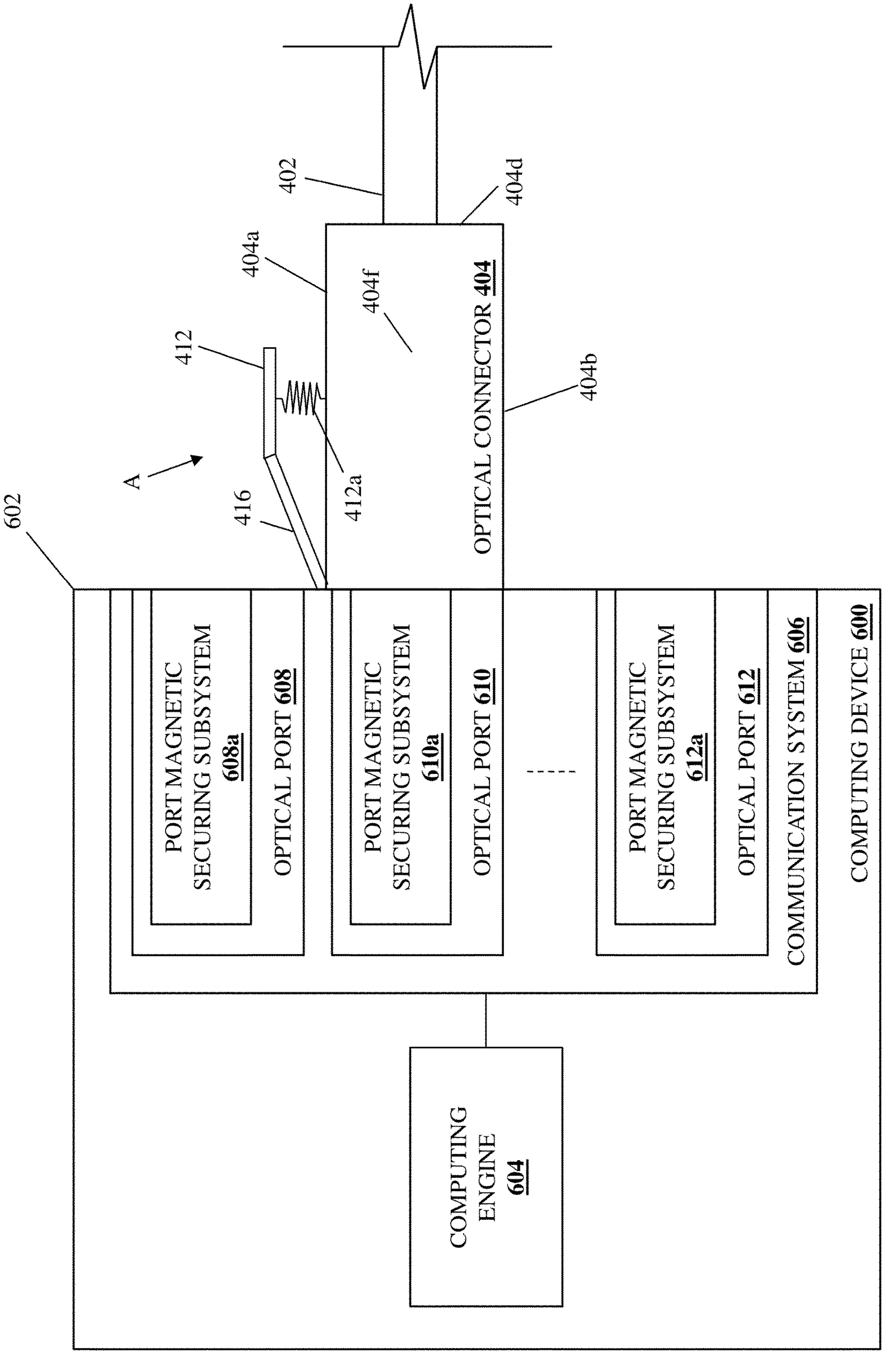
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 10B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PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
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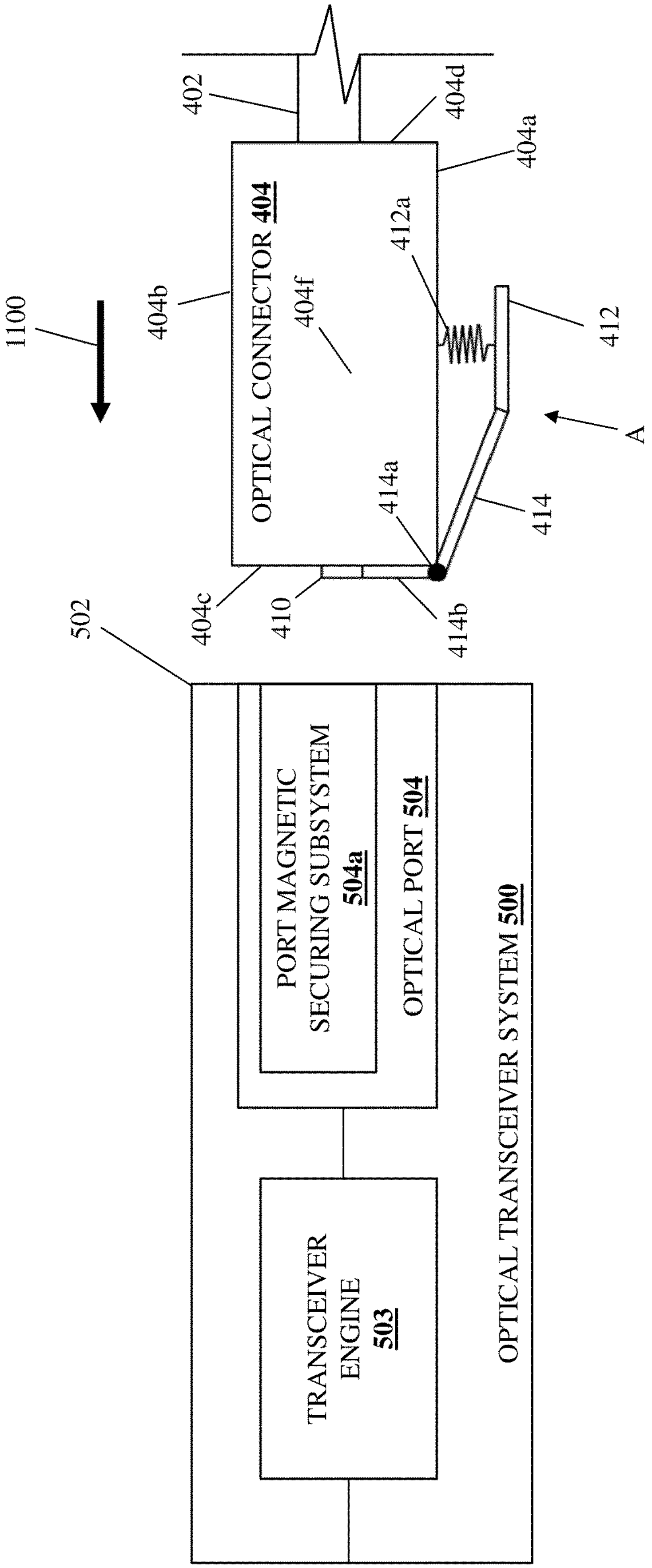
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 10C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A
414a
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404c
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414b
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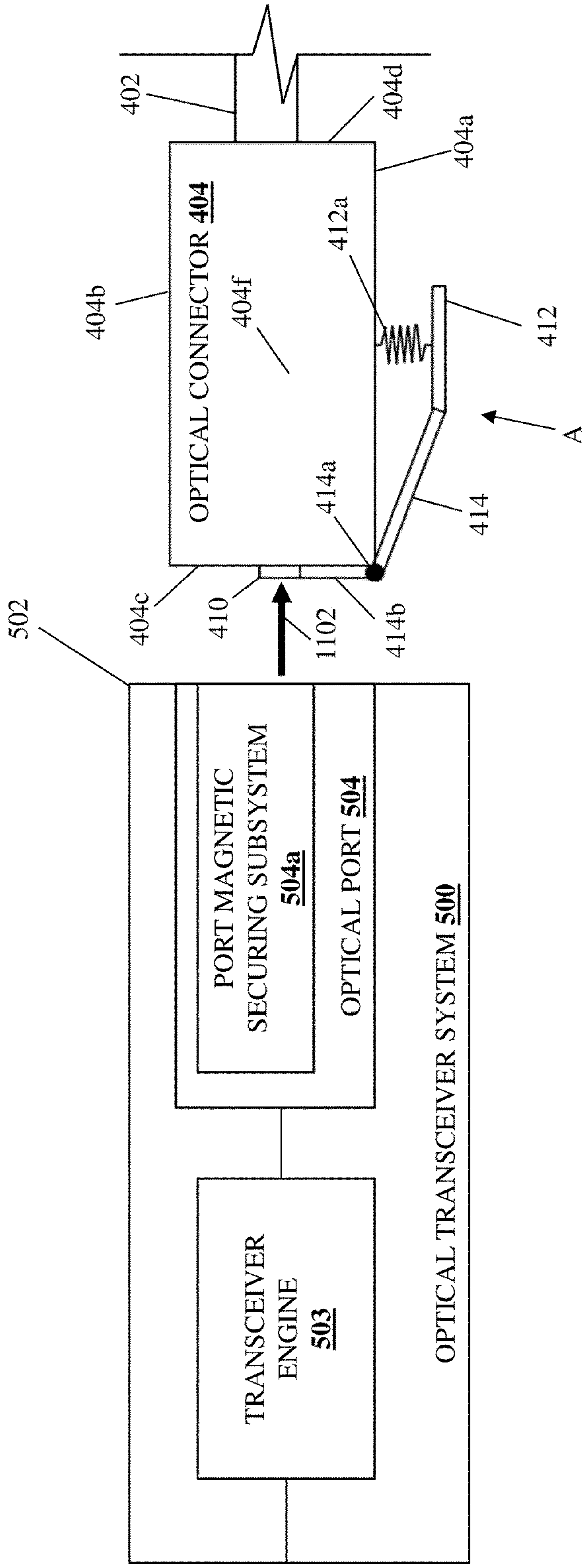
PORT MAGNETIC SECURING SUBSYSTEM 504a
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER ENGINE 503

FIG. 11A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404c
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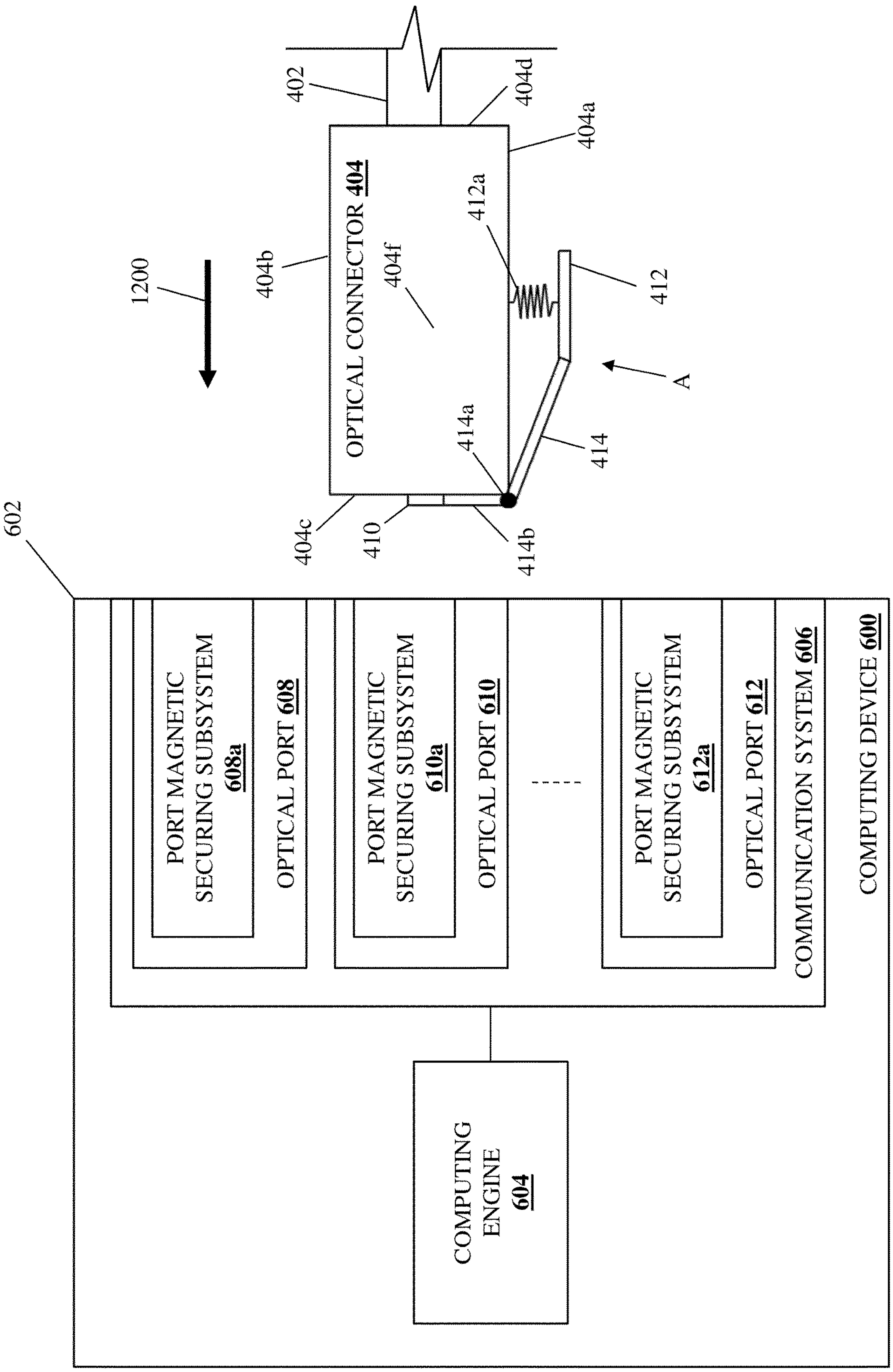
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER ENGINE 503

FIG. 11B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404b
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A
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414
404c
410
414b
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OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
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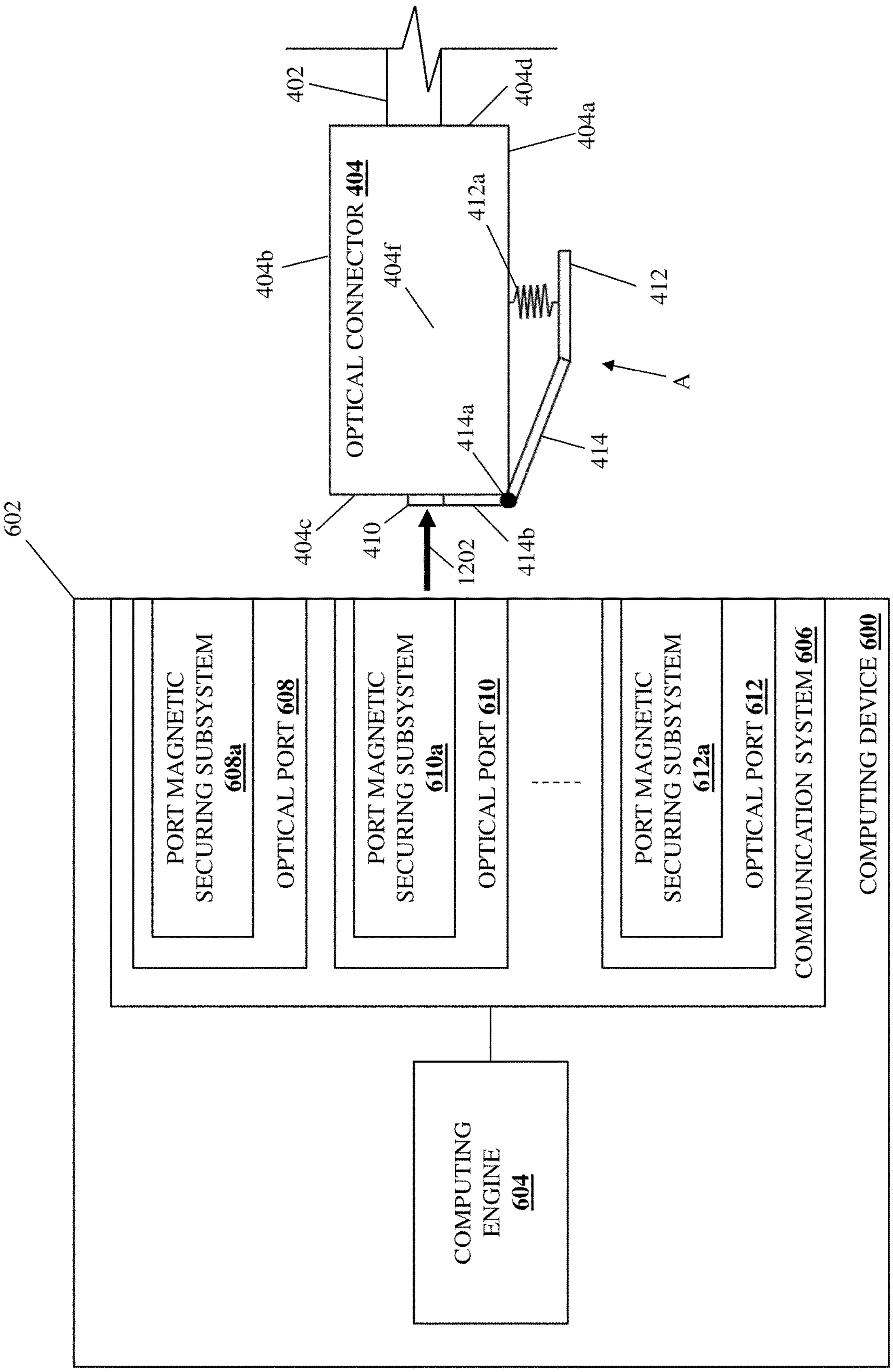
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 12A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404c
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1202
414b
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OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
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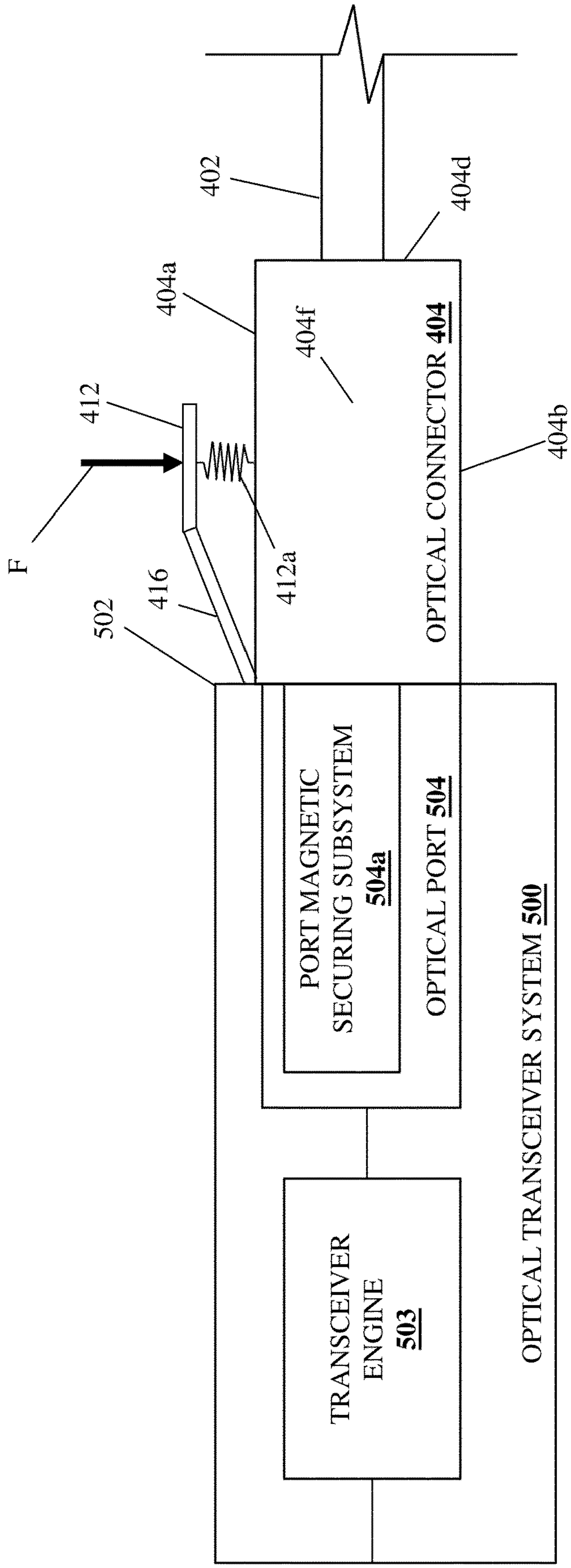
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 12B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404b
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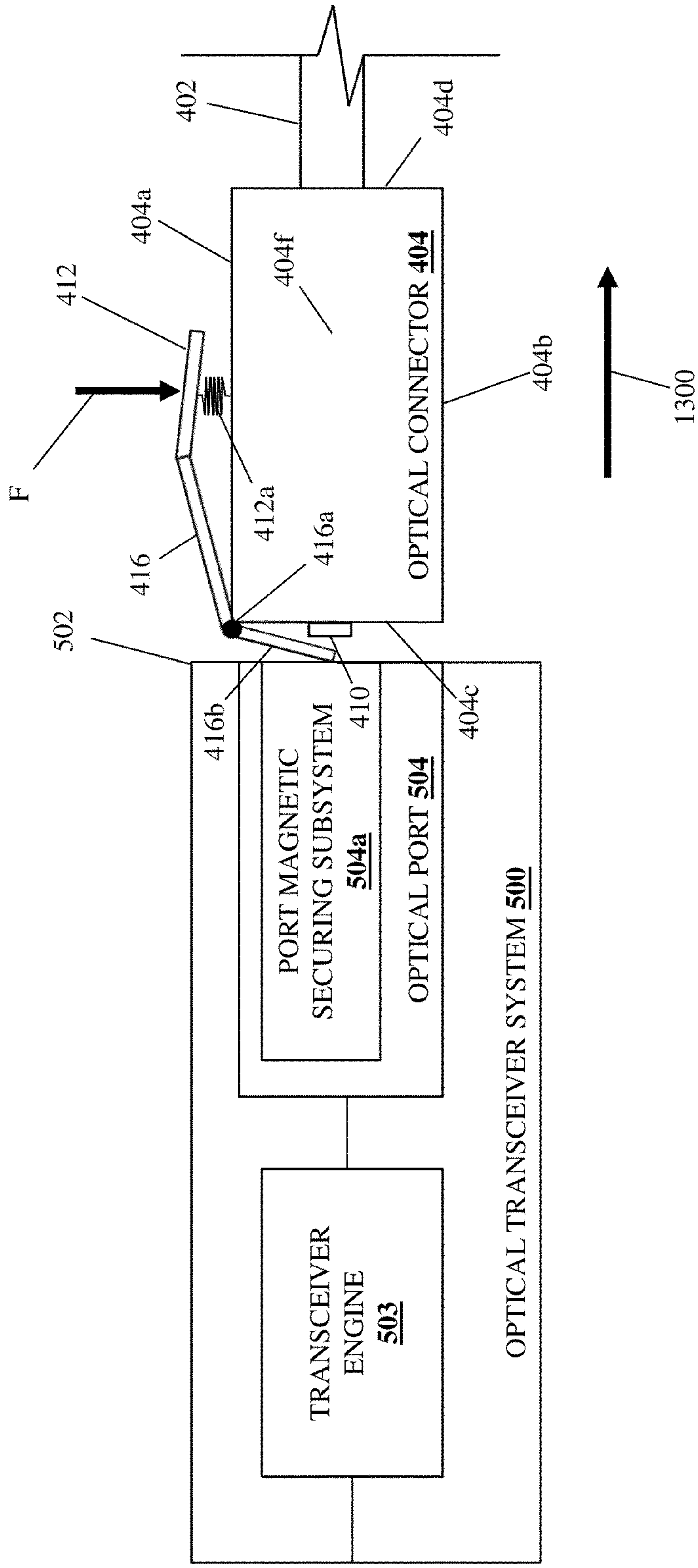
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER ENGINE 503

FIG. 13A



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1300
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412a
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416a
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416b
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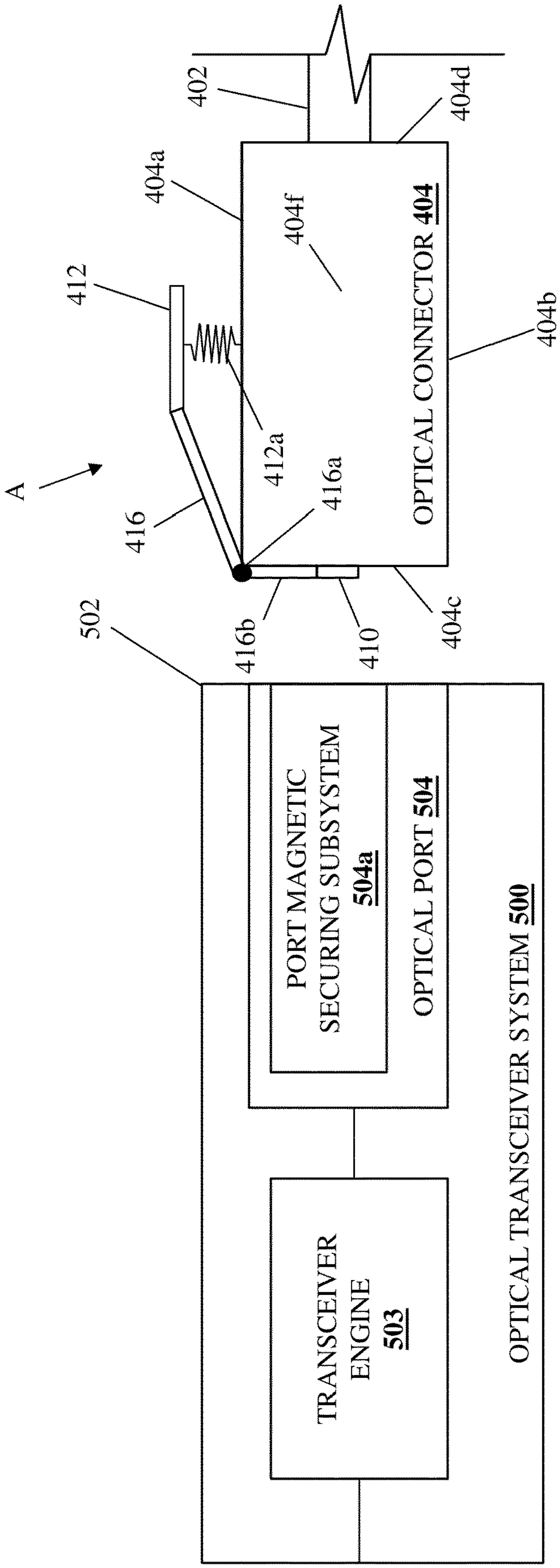
404c
PORT MAGNETIC SECURING SUBSYSTEM 504a
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER ENGINE 503

FIG. 13B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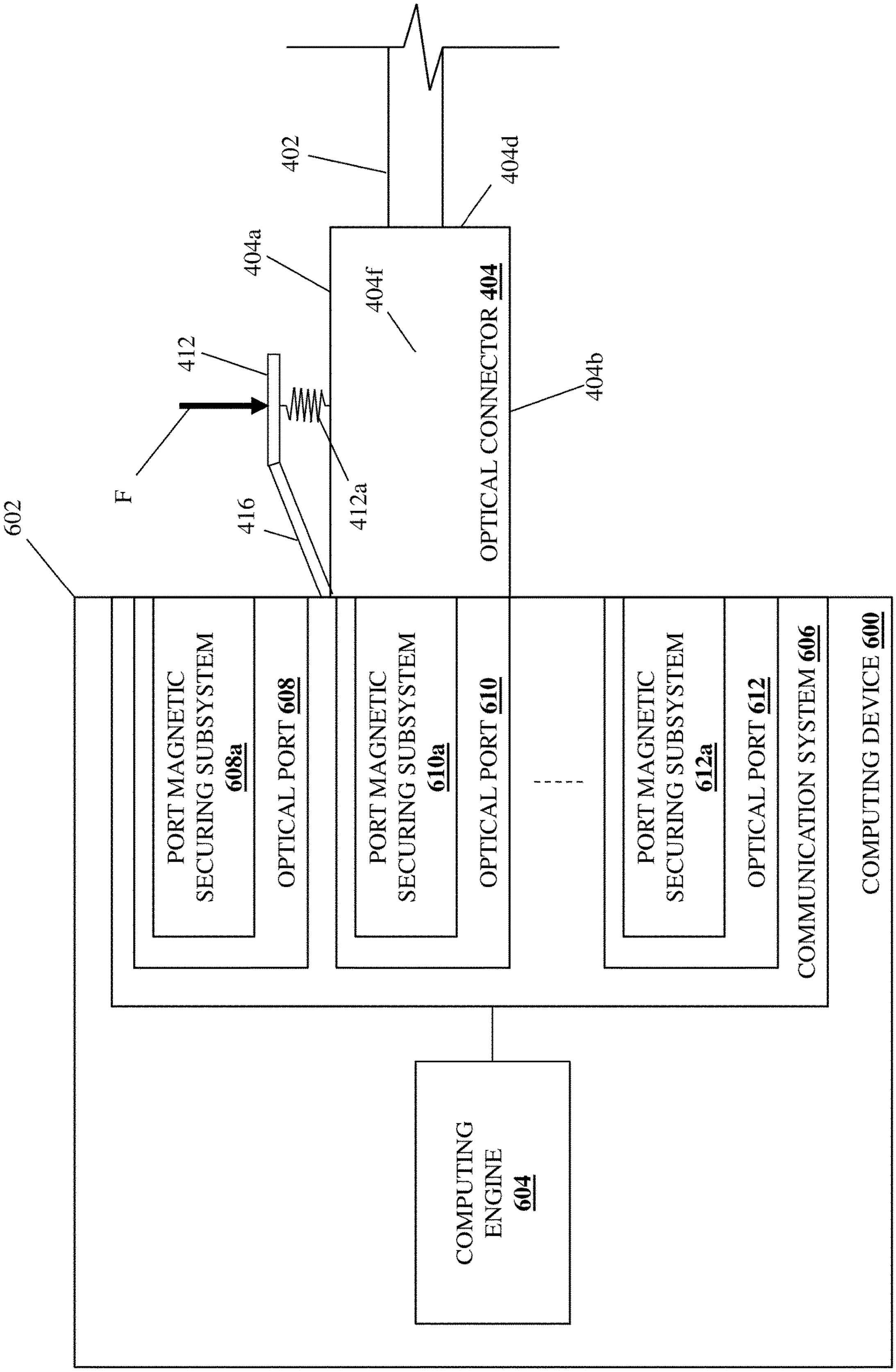
OPTICAL PORT 504
OPTICAL TRANSCEIVER SYSTEM 500
TRANSCEIVER ENGINE 503

FIG. 13C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PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
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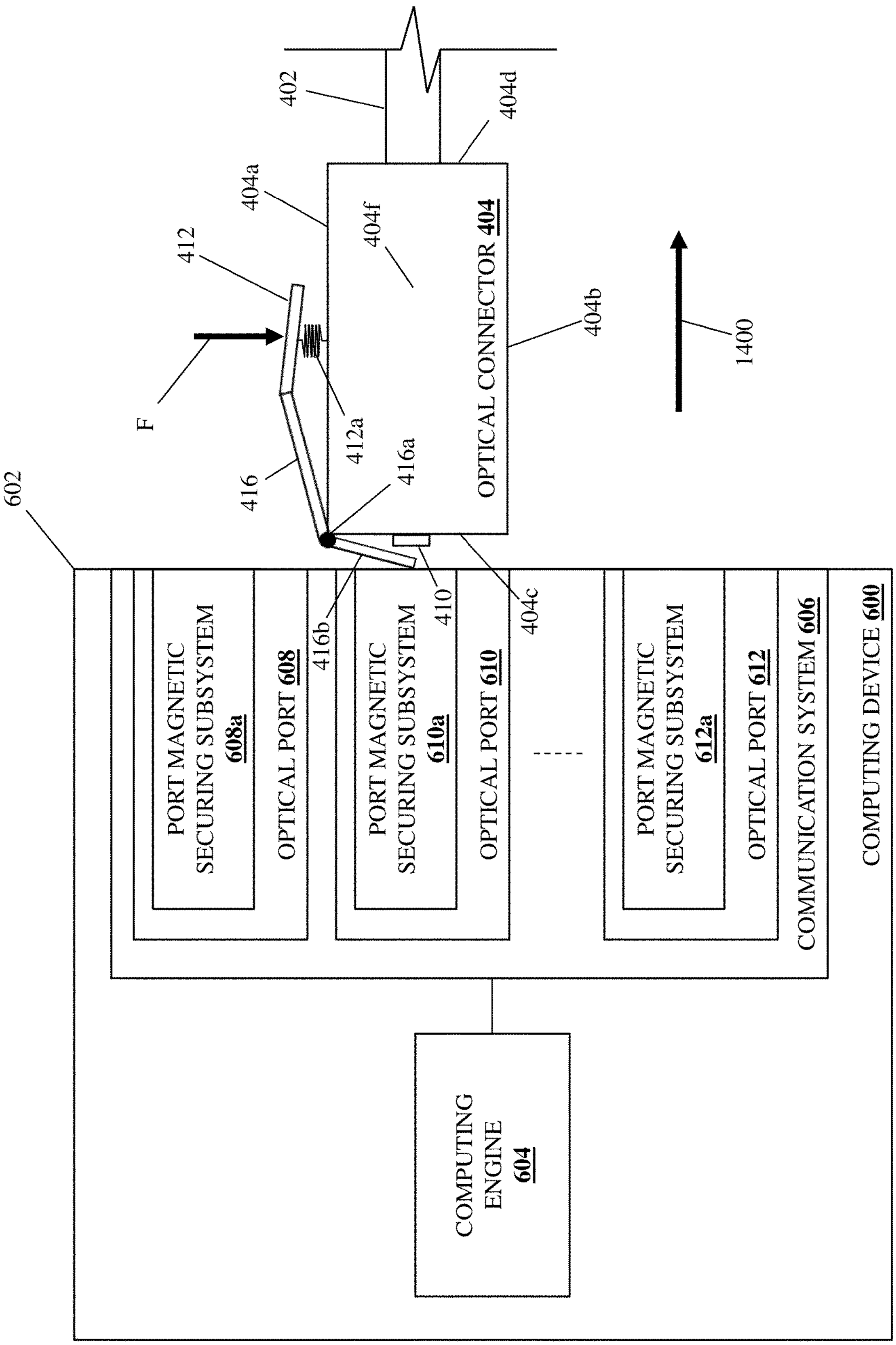
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 14A


402
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OPTICAL CONNECTOR 404
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PORT MAGNETIC SECURING SUBSYSTEM 608a
OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
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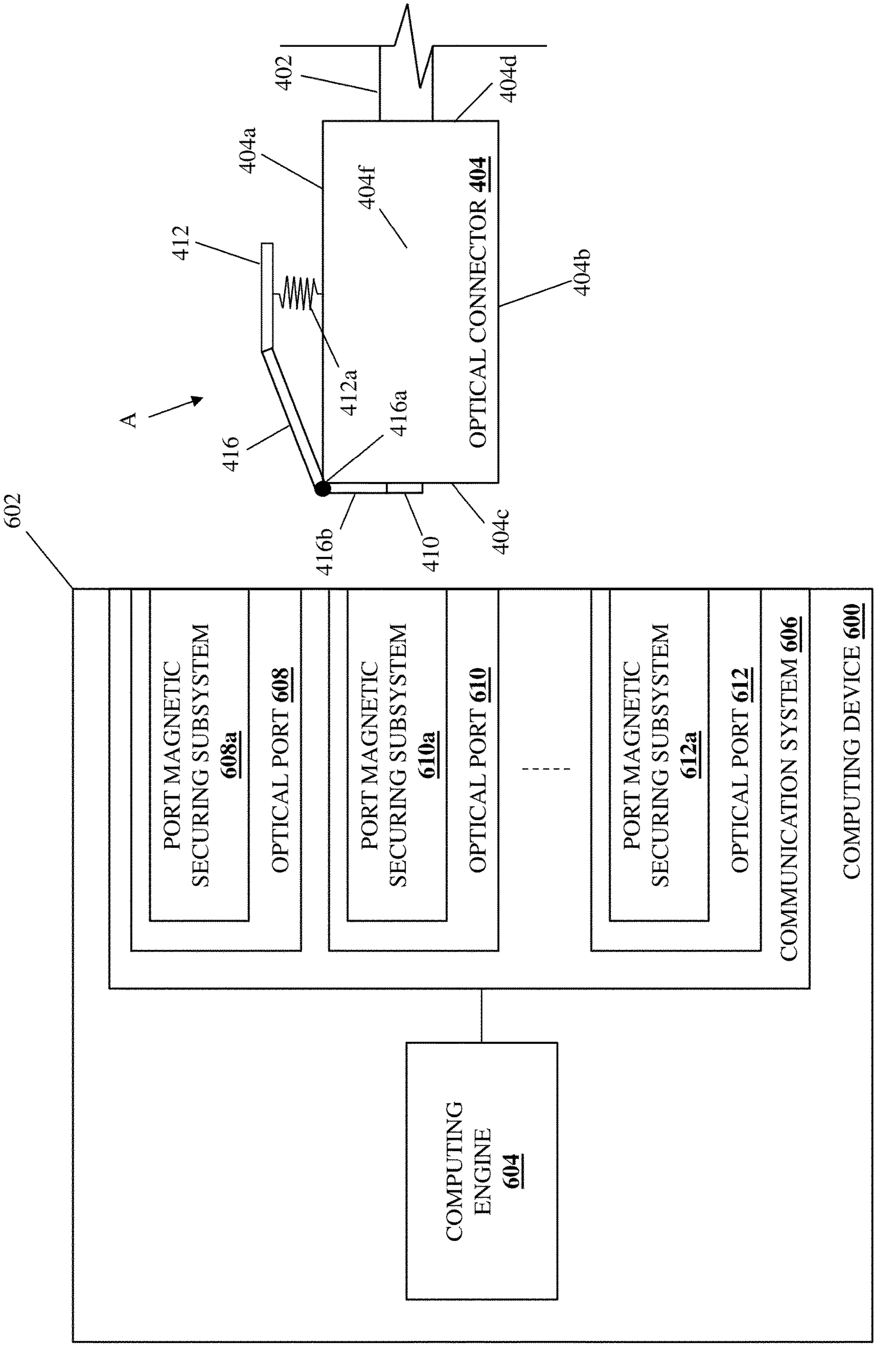
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 14B


602
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A
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416b
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OPTICAL PORT 608
PORT MAGNETIC SECURING SUBSYSTEM 610a
OPTICAL PORT 610
PORT MAGNETIC SECURING SUBSYSTEM 612a
OPTICAL PORT 612
COMMUNICATION SYSTEM 606
COMPUTING DEVICE 600
COMPUTING ENGINE 604

FIG. 14C

OPTICAL PORT/CONNECTOR MAGNETIC SECURING/RELEASE SYSTEM

BACKGROUND

[0001] The present disclosure relates generally to information handling systems, and more particularly to magnetically securing and releasing an optical connector to and from an optical port on an information handling system.

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

[0003] Information handling systems such as, for example, switch devices, other networking devices, storage systems, server devices, and/or other computing devices known in the art, sometimes utilize optical data transmission systems in order to transmit data. For example, switch devices often utilize fiber optic ports (e.g., fiber optic ports integrated with the switch device, fiber optic ports provided on fiber optic transceiver systems coupled to the switch device, etc.) in order to optically transmit data via fiber optic cables connected to those fiber optic ports via fiber optic connectors. However, the conventional securing of such fiber optic connectors to fiber optic ports can raise some issues.

[0004] For example, conventional fiber optic cables may utilize Lucent Connectors (LCs) including a plastic housing with a latch that enables "push/pull" functionality with a corresponding fiber optic port, Standard Connectors (SCs) that are similar to LCs but include a locking tab in place of the latch, Ferrule Core (FC) connectors that are threaded to screw on to a corresponding fiber optic port, Straight Tip (ST) connectors that are similar to FC connectors but that replace the threads with a locking mechanism provided by a plug and a socket that are locked by a half-twist bayonet lock, Multi-fiber Push-On (MPO) connectors used to connect ribbon cables with multiple fibers to a corresponding fiber optic port, as well as other conventional fiber optic connectors known in the art.

[0005] As will be appreciated by one of skill in the art, such conventional fiber optic connectors include a relatively high number of components that can increase their cost, and many of which move relative to each other and that cause wear that requires their replacement over time. One example of such components are those included in the conventional alignment mechanisms that are used with conventional fiber optic connectors to ensure they are oriented correctly with respect to an fiber optic port in order to operate properly, and those conventional alignment mechanisms suffer from additional issues as they often make it difficult to properly align those fiber optic connectors for engagement with corresponding fiber optic ports, while requiring careful attention so as to not damage the fiber optic connector or the fiber optic port.

[0006] Accordingly, it would be desirable to provide a fiber optic port/connector securing/release system that addresses the issues discussed above.

SUMMARY

[0007] According to one embodiment, an Information Handling System (IHS) includes a chassis; a processing system that is housed in the chassis; an optical port that is included on the chassis and that is coupled to the processing system; a port magnetic securing subsystem that is included on the optical port; an optical cable; an optical connector that is included on the optical cable; a connector magnetic securing subsystem that is included on the optical connector and that is magnetically connected to the port magnetic securing subsystem to secure the optical connector to the optical port; and a mechanical release device that is coupled to the optical connector and that is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a schematic view illustrating an embodiment of an Information Handling System (IHS).

[0009] FIG. **2** is a schematic view illustrating an embodiment of an optical cable system that may be provided according to the teachings of the present disclosure.

[0010] FIG. **3**A is a schematic view illustrating an embodiment of an optical connector that may be included on the optical cable system of FIG. **2**.

[0011] FIG. **3**B is a schematic view illustrating an embodiment of an optical connector that may be included on the optical cable system of FIG. **2**.

[0012] FIG. **4**A is a perspective view illustrating an embodiment of the optical cable system of FIG. **2** with its mechanical release device in a securing orientation.

[0013] FIG. **4**B is a side view illustrating an embodiment of the optical cable system of FIG. **2** with its mechanical release device in a securing orientation.

[0014] FIG. **4**C is a side view illustrating an embodiment of the optical cable system of FIG. **2** with its mechanical release device in a release orientation.

[0015] FIG. **5** is a schematic view illustrating an embodiment of an optical transceiver system that may be used with the optical cable system of FIG. **2**.

[0016] FIG. **6** is a schematic view illustrating an embodiment of a computing device that may be used with the optical cable system of FIG. **2**.

[0017] FIG. **7**A is a schematic view illustrating an embodiment of an optical port that may be included on the optical transceiver system of FIG. **5** or the computing device of FIG. **6**.

[0018] FIG. 7B is a schematic view illustrating an embodiment of an optical port that may be included on the optical transceiver system of FIG. 5 or the computing device of FIG. 6.

[0019] FIG. 8 is a flow chart illustrating an embodiment of a method for magnetically securing and releasing an optical connector to and from an optical port.

[0020] FIG. 9A is a schematic view illustrating an embodiment of an optical connector on the optical cable system of FIGS. 4A-4C being moved adjacent the optical transceiver system of FIG. 5 while oriented with an optical port on the optical transceiver system.

[0021] FIG. 9B is a schematic view illustrating an embodiment of the optical connector on the optical cable system of FIGS. 4A-4C magnetically interacting with the optical port on the optical transceiver system of FIG. 5.

[0022] FIG. 9C is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C connected to the optical transceiver system of FIG. 5.

[0023] FIG. 10A is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C being connected to the computing device of FIG. 6.

[0024] FIG. 10B is a schematic view illustrating an embodiment of an optical connector on the optical cable system of FIGS. 4A-4C magnetically interacting with an optical port on the computing device of FIG. 6.

[0025] FIG. 10C is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C connected to the computing device of FIG. 6.

[0026] FIG. 11A is a schematic view illustrating an embodiment of an optical connector on the optical cable system of FIGS. 4A-4C being moved adjacent the optical transceiver system of FIG. 5 while misoriented with an optical port on the optical transceiver system.

[0027] FIG. 11B is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C being prevented from being connected to the optical transceiver system of FIG. 5 when the optical connector on the optical cable system is misoriented with the optical port on the optical transceiver system.

[0028] FIG. 12A is a schematic view illustrating an embodiment of an optical connector on the optical cable system of FIGS. 4A-4C being moved adjacent the computing device of FIG. 6 while misoriented with an optical port on the computing device.

[0029] FIG. 12B is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C being prevented from being connected to the computing device of FIG. 6 when the optical connector on the optical cable system is misoriented with the optical port on the computing device.

[0030] FIG. 13A is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C being disconnected from the optical transceiver system of FIG. 5.

[0031] FIG. 13B is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C disconnected from the optical transceiver system of FIG. 5.

[0032] FIG. 13C is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C disconnected from the optical transceiver system of FIG. 5.

[0033] FIG. 14A is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C being disconnected from the computing device of FIG. 6.

[0034] FIG. 14B is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C disconnected from the computing device of FIG. 6.

[0035] FIG. 14C is a schematic view illustrating an embodiment of the optical cable system of FIGS. 4A-4C disconnected from the computing device of FIG. 6.

DETAILED DESCRIPTION

[0036] For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0037] In one embodiment, IHS 100, FIG. 1, includes a processor 102, which is connected to a bus 104. Bus 104 serves as a connection between processor 102 and other components of IHS 100. An input device 106 is coupled to processor 102 to provide input to processor 102. Examples of input devices may include keyboards, touchscreens, pointing devices such as mouses, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device 108, which is coupled to processor 102. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS 100 further includes a display 110, which is coupled to processor 102 by a video controller 112. A system memory 114 is coupled to processor 102 to provide the processor with fast storage to facilitate execution of computer programs by processor 102. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis 116 houses some or all of the components of IHS 100. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor 102 to facilitate interconnection between the components and the processor 102.

[0038] Referring now to FIG. 2, an embodiment of an optical cable system 200 is illustrated that may be provided according to the teachings of the present disclosure. In the illustrated embodiment, the optical cable system 200 includes an optical cable **202** that one of skill in the art in possession of the present disclosure will appreciate may be provided by any of a variety of fiber optic cables known in the art. As illustrated, an optical connector **204** is included on an end of the optical cable **202**, and one of skill in the art in possession of the present disclosure will appreciate how at least one optical connector (and a plurality of optical connectors in the case of a "breakout" type cable) may be provided on opposing, unillustrated ends of the optical cable **202** while remaining within the scope of the present disclosure as well. As illustrated, a connector magnetic securing subsystem **204***a* is included on the optical connector **204**, and as described below may be configured in a variety of manners to secure the optical connector **204** to an optical port. Furthermore, a mechanical release device **206** is included on the optical connector **204**, and as described below may be configured in a variety of manners to mechanically disconnect the connector magnetic securing subsystem **204***a* from an optical port to which it is connected in order to release the optical connector **204** from the optical port. However, while some specific examples of the connector magnetic securing subsystem **204***a* and the mechanical release device **206** are provided below, one of skill in the art in possession of the present disclosure will appreciate how the connector magnetic securing subsystem and the mechanical release device of the present disclosure may be configured in a variety of manners in order to provide the functionality described below.

[0039] With reference to FIG. **3**A, an embodiment of an optical connector **300** is illustrated that may provide the optical connector **204** discussed above with reference to FIG. **2**, but with the mechanical release device **206** omitted for clarity. In the illustrated embodiment, the optical connector **300** includes a base **302** having an optical conduit subsystem **304** that one of skill in the art in possession of the present disclosure will appreciate may include fiber ferrule (s) (e.g., which are coupled to optical fibers in an attached optical cable) and/or other fiber optic components known in the art. As will be appreciated by one of skill in the art in possession of the present disclosure, the base **302** and the optical conduit subsystem **304** of the optical connector **300** may be similar to the base and optical conduit subsystem included on conventional fiber optic connectors (e.g., the LCs, SCs, FC connectors, ST connectors, and MPO connectors discussed above, as well as other fiber optic connectors known in the art), but with the optical port alignment and connection components on those conventional fiber optic connectors replaced by the optical port alignment and connection components described below. Furthermore, one of skill in the art in possession of the present disclosure will appreciate how the optical connectors of the present disclosure eliminate many components utilized in the conventional optical connectors referenced above (e.g., alignment components, latches/pull tabs used to disconnect conventional optical connectors from their optical port, etc.), thus reducing material costs, manufacturing costs, points of failure, and overall complexity.

[0040] As illustrated, the base **302** of the optical connector **300** may include a connector magnetic securing subsystem that provides the connector magnetic securing subsystem **204***a* on the optical connector **204** discussed above with reference to FIG. **2**. In the examples illustrated and described below, the connector magnetic securing subsystem on the optical connector **300** includes a pair of magnets **306** and **308** that are located on the base **302** on opposite sides of the optical conduit subsystem **304**. As will be appreciated by one of skill in the art in possession of the present disclosure, the use of magnets with the optical connector of the present disclosure will not require any added shielding (as would be required if provided on an electrical connector) due to the use of light to optically transmit data.

[0041] In a specific example, the magnets **306** and **308** may be provided by neodymium magnets that one of skill in the art in possession of the present disclosure will appreciate are relatively high strength magnets with relatively high temperature resistance that allow for the use of relatively small magnets in relatively high temperature environments like datacenters, although one of skill in the art in possession of the present disclosure will appreciate how other types of magnets may fall within the scope of the present disclosure as well. However, while a pair of specific types of magnets in a specific configuration is illustrated, one of skill in the art in possession of the present disclosure will appreciate how the connector magnetic securing subsystem on the optical connectors of the present disclosure may include a single magnet, or more than two magnets, in a variety of configurations while remaining within the scope of the present disclosure as well.

[0042] Furthermore, with reference to FIG. **3**B, in some embodiments the connector magnetic securing subsystem on the optical connector **300** may be provided with a connector "keying" subsystem that is configured to ensure that the optical connector **300** is properly oriented with respect to a corresponding optical port in order to connect the optical connector **300** to that optical port. In the specific example illustrated in FIG. **3**B, the connector keying system is provided by a pair of magnets **306***a* and **308***a* that are located on the base **302** on opposite sides of the optical conduit subsystem **304**, with the magnet **306***a* provided with its north pole end exposed on the surface of the optical connector **300** (i.e., as indicated by the vertical lines on element **306***a*), and with the magnet **308***a* provided with its south pole end exposed on the surface of the optical connector **300** (as indicated by the horizontal lines on element **308***a*). As described below, the connector keying subsystem on the optical connector **300** may be configured to magnetically interact with a port keying subsystem on an optical port, and while a specific port/connector keying system is described, one of skill in the art in possession of the present disclosure will appreciate how the port/connector keying system of the present disclosure may be provided in a variety of manners while remaining within the scope of the present disclosure as well.

[0043] Furthermore, while the specific example provided above and discussed below includes magnets on the connector magnetic securing subsystem of the optical connector **300**, one of skill in the art in possession of the present disclosure will appreciate how the connector magnetic securing subsystem of the optical connector **300** may instead be provided by a ferromagnetic material (e.g., Cobalt, Iron, Iron alloys (e.g., steel), Nickel, Gadolinium, Dysprosium, etc.) that is configured to magnetically interact with magnets included in a port magnetic securing subsystem on an optical port in other embodiments of the present disclosure that are discussed in further detail below. As such, in some embodiments, the material used to provide the front surface **404***c* of the optical connector **404** may be provided by a non-ferromagnetic material (e.g., any of a variety of plastic materials or other non-magnetic materials known in the art) to enable at least some of the functionality described below without interfering with the magnet/ferromagnetic material or magnet/magnet interactions described herein.

[0044] With reference to FIG. **4**A, an embodiment of an optical cable system **400** is illustrated that may provide the optical cable system **200** discussed above with reference to FIG. **2**. In the illustrated embodiment, the optical cable system **400** includes an optical cable **402** that may provide the optical cable **202** discussed above with reference to FIG. **2**. As illustrated, an optical connector **404** that may provide the optical cable **204** discussed above with reference to FIG. **2** or the optical cable **300** discussed above with reference to FIGS. **3**A and **3**B is included on an end of the optical cable **402**, and similarly as described above, at least one optical connector (and a plurality of optical connectors in the case of a "breakout" type cable) may be provided on opposing, unillustrated ends of the optical cable **402** while remaining within the scope of the present disclosure as well. In the illustrated embodiment, the optical connector **404** includes a top surface **404***a*, a bottom surface **404***b* that is located opposite the optical connector **404** from the top surface **404***a*, a front surface **404***c* that extends between the top surface **404***a* and the bottom surface **404***b*, a rear surface **404***d* that is located opposite the optical connector **404** from the front surface **404***c* and that extends between the top surface **404***a* and the bottom surface **404***b*, and a pair of side surfaces **404***e* and **404***f* that are located opposite the optical connector **404** from each other and that each extend between the top surface **404***a*, the bottom surface **404***b*, the front surface **404***c*, and the bottom surface **404***d*.

[0045] As will be appreciated by one of skill in the art in possession of the present disclosure, the optical cable **402** extends from the rear surface **404***d* of the optical connector **404**, and may include optical fiber(s) that are coupled to an optical conduit subsystem **406** (e.g., fiber ferrule(s) and other optical components known in the art) that is included in the optical connector **404** and a portion of which is accessible on the front surface **404***c* of the optical connector **404**. In the illustrated embodiments, a connector magnetic securing subsystem is included on the optical connector **404**, and in the illustrated examples is provided by a pair of magnets **408** and **410** that are located on the front surface **404***c* of the optical connector **404** and on opposite sides of the optical conduit subsystem **406**. As will be appreciated by one of skill in the art in possession of the present disclosure, the magnets **408***a* and **410** that provide the connector magnetic securing subsystem on the optical connector **404** may be provided by the magnets **306** and **308** discussed above with reference to FIG. **3**A, or the magnets **306***a* and **308***a* discussed above with reference to FIG. **3**B.

[0046] Furthermore, a mechanical release device is included on the optical connector **404**, and in the illustrated example is provided by an actuator element **412** that is resiliently coupled to the top surface **404***a* of the optical connector **404** by a spring device **412***a*, along with a pair of arm members **414** and **416** that extend from the actuator element **412** (and away from each other along their length) to a moveable coupling on the optical connector **404** that is provided by respective hinges **414***a* and **416***a* that are spaced apart on the corner of the optical connector **404** between the top surface **404***a* and the front surface **404***c*, with each arm member **414** and **416** including a port engagement portion **414***b* and **416***b*, respectively, that extends from their respective hinges **414***a* and **416***a* substantially parallel to each other and adjacent the front surface **404***c* of the optical connector **404**.

[0047] As illustrated in FIGS. **4**B and **4**C, the mechanical release device on the optical connector **404** is moveable between a securing orientation A (illustrated in FIG. **4**B) and a release orientation B (illustrated in FIG. **4**C). For example, in response to a force F applied to the actuator element **412** (e.g., via a finger of the user of the optical cable system **400**) that is sufficient to compress the spring device **412***a*, the arm members **414** and **416** may rotate in a direction R about their respective hinges **414***a* and **416***a* such that their respective port engagement portions **414***b* and **416***b* move away from the front surface **404***c* of the optical connector **404**. However, while a specific mechanism for the mechanical release device on the optical connector **404** is illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how the mechanical release functionality of the present disclosure may be provided by a variety of mechanisms that will fall within the scope of the present disclosure as well. Furthermore, while several specific examples and embodiments of optical cable systems have been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that the optical cable system of the present disclosure may include a variety of components and component configurations for providing conventional optical cable functionality, as well as the optical port/connector magnetic secure/release functionality discussed below, while remaining within the scope of the present disclosure as well.

[0048] Referring now to FIG. **5**, an embodiment of an optical transceiver system **500** is illustrated that may be used with the optical cable system **200** of FIG. **2**. In an embodiment, the optical transceiver system **500** may be provided by the IHS **100** discussed above with reference to FIG. **1** and/or may include some or all of the components of the IHS **100**, and in specific examples may be provided by Small Form-factor Pluggable (SFP) optical transceivers, enhanced SFP (SFP+) optical transceivers, 10-gigabit SFP (XFP) optical transceivers, Quad SFP (QSFP) optical transceivers, and/or other optical transceivers that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, while illustrated and discussed as being provided by specific optical transceivers, one of skill in the art in possession of the present disclosure will recognize that the functionality of the optical transceiver system **500** discussed below may be provided by other devices that are configured to operate similarly as the optical transceiver system **500** discussed below.

[0049] In the illustrated embodiment, the optical transceiver system **500** includes a chassis **502** that houses the components of the optical transceiver system **500**, only some of which are illustrated and described below. For example, the chassis **502** may house a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a transceiver engine

503 that is configured to perform the functionality of the transceiver engines and/or optical transceiver systems discussed below.

[0050] The chassis **502** may also house an optical port **504** that is accessible via a surface of the chassis **502**, that is coupled to the transceiver engine **503** (e.g., via a coupling between the optical port **504** and the processing system), and that may be configured to couple to any of the optical connectors described herein. As illustrated, a port magnetic securing subsystem **504***a* is included on the optical connector **504**, and as described below may be configured in a variety of manners to secure any of the optical connectors described herein to the optical port **504**. Furthermore, a device connector **506** is included on the chassis **502** and coupled to the transceiver engine **503** (e.g., via a coupling between the device connector **506** and the processing system), and may be provided by switch port connectors and/or other device port connectors that one of skill in the art in possession of the present disclosure would recognize as being configured to connect the optical transceiver system **500** to a conventional switch or other conventional computing devices known in the art. However, while a specific optical transceiver system **500** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that optical transceiver systems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the optical transceiver system **500**) may include a variety of components and/or component configurations for providing conventional optical transceiver functionality, as well as the optical port/connector magnetic secure/release functionality discussed below, while remaining within the scope of the present disclosure as well.

[0051] Referring now to FIG. **6**, an embodiment of computing device **600** is illustrated that may be used with the optical cable system **200** of FIG. **2**. In an embodiment, the computing device **600** may be provided by the IHS **100** discussed above with reference to FIG. **1** and/or may include some or all of the components of the IHS **100**, and in specific examples may be provided by a switch device or other networking devices, storage systems, server devices, and/or other computing devices that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, while illustrated and discussed as being provided by specific devices, one of skill in the art in possession of the present disclosure will recognize that the functionality of the computing device **600** discussed below may be provided by other devices that are configured to operate similarly as the computing device **600** discussed below.

[0052] In the illustrated embodiment, the computing device **600** includes a chassis **602** that houses the components of the computing device **600**, only some of which are illustrated and described below. For example, the chassis **602** may house a processing system (not illustrated, but which may be similar to the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may be similar to the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a computing engine **604** that is configured to perform the functionality of the computing engines and/or computing devices discussed below.

[0053] In the illustrated embodiment, the chassis **602** houses a communication system **606** that includes plurality of optical ports **608**, **610**, and up to 612 that are each accessible via a surface of the chassis **602**, that are each coupled to the computing engine **604** (e.g., via a coupling between that optical port and the processing system), and that may each be configured to couple to any of the optical connectors described herein. However, while one of skill in the art in possession of the present disclosure will appreciate that the computing device **600** is provided by a switch device having a plurality of optical ports in the examples provided herein, other computing devices may include fewer optical ports (including only a single optical port) while remaining within the scope of the present disclosure as well.

[0054] As illustrated, a port magnetic securing subsystem is included on each of the optical ports in the communication system **606**, with a port magnetic securing subsystem **608***a* included on the optical port **608**, a port magnetic securing subsystem **610***a* included on the optical port **610**, and up to a port magnetic securing subsystem **612***a* included on the optical port **612**. As described below, any of the port magnetic securing subsystems **608***a*-**612***a* may be configured in a variety of manners to secure any of the optical connectors described herein to their optical port **608**-**612**, respectively. However, while a specific computing device **600** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that computing devices (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the computing device **600**) may include a variety of components and/or component configurations for providing conventional computing functionality, as well as the optical port/connector magnetic secure/release functionality discussed below, while remaining within the scope of the present disclosure as well.

[0055] With reference to FIG. 7A, an embodiment of an optical port **700** is illustrated that may provide any of the optical port **504** discussed above with reference to FIG. **5**, or the optical ports **608**-**612** discussed above with reference to FIG. **6**. In the illustrated embodiment, the optical port **700** includes a base **702** having an optical conduit subsystem **704** that one of skill in the art in possession of the present disclosure will appreciate may include fiber ferrule receiver (s) and/or other fiber optic components known in the art. As will be appreciated by one of skill in the art in possession of the present disclosure, the base **702** and the optical conduit subsystem **704** of the optical port **700***n* may be similar to conventional fiber optic ports that are configured to connect to conventional fiber optic connectors (e.g., the LCs, SCs, FC connectors, ST connectors, and MPO connectors discussed above, as well as other fiber optic connectors known in the art), but with the optical connector alignment and connection components on those conventional fiber optic ports replaced by the optical connector alignment and connection components described below.

[0056] As illustrated, the base **702** of the optical port **700** may include a port magnetic securing subsystem that provides any of the port magnetic securing subsystem **504***a* on the optical port **504** discussed above with reference to FIG. **5**, or the port magnetic securing subsystems **608***a*-**612***a* on the optical ports **608**-**612** discussed above with reference to FIG. **6**. In the examples illustrated and described below, the port magnetic securing subsystem on the optical port **700** may include a pair of magnets **706** and **708** that are located on the base **702** on opposite sides of the optical conduit subsystem **704**. As will be appreciated by one of skill in the art in possession of the present disclosure, the use of magnets with the optical port of the present disclosure will not require any added shielding (as would be required if provided on an electrical port) due to the use of light to optically transmit data.

[0057] In a specific example, the magnets **706** and **708** may be provided by neodymium magnets that one of skill in the art in possession of the present disclosure will appreciate are relatively high strength magnets with relatively high temperature resistance that allow for the use of relatively small magnets in relatively high temperature environments like datacenters, although one of skill in the art in possession of the present disclosure will appreciate how other types of magnets may fall within the scope of the present disclosure as well. However, while a pair of specific types of magnets in a specific configuration is illustrated, one of skill in the art in possession of the present disclosure will appreciate how the port magnetic securing subsystem on the optical ports of the present disclosure may include a single magnet, or more than two magnets, in a variety of configurations while remaining within the scope of the present disclosure as well.

[0058] Furthermore, with reference to FIG. **7**B, in some embodiments the port magnetic securing subsystem on the optical port **700** may be provided with a port "keying" subsystem that is configured to ensure that an optical connector provided according to the teachings of the present disclosure is properly oriented with respect to the optical port **700** in order to connect that optical connector to the optical port **700**. In the specific example illustrated in FIG. **7**B, the port keying system is provided by a pair of magnets ***706a*** and ***708a*** that are located on the base **702** on opposite sides of the optical conduit subsystem **704**, with the magnet ***706a*** provided with its north pole end exposed on the surface of the optical port **700** (i.e., as indicated by the vertical lines on element ***706a***), and with the magnet ***708a*** provided with its south pole end exposed on the surface of the optical port **700** (as indicated by the horizontal lines on element ***708a***). As described below, the port keying subsystem on the optical port **700** may be configured to magnetically interact with a connector keying subsystem on an optical connector, and while a specific port/connector keying system is described, one of skill in the art in possession of the present disclosure will appreciate how the port/connector keying system of the present disclosure may be provided in a variety of manners while remaining within the scope of the present disclosure as well.

[0059] Furthermore, while the specific example provided above and discussed below includes magnets on the port magnetic securing subsystem of the optical port **700**, one of skill in the art in possession of the present disclosure will appreciate how the port magnetic securing subsystem of the optical port **700** may instead be provided by a ferromagnetic material (e.g., Cobalt, Iron, Iron alloys (e.g., steel), Nickel, Gadolinium, Dysprosium, etc.) that is configured to magnetically interact with magnets included in a connector magnetic securing subsystem on an optical connector in other embodiments of the present disclosure that are discussed in further detail below. As such, in some embodiments, the material used to provide the front surface of the optical port **700** upon which the port magnetic securing subsystem is located may be provided by a non-ferromagnetic material (e.g., any of a variety of plastic materials or other non-magnetic materials known in the art) to enable at least some of the functionality described below without interfering with the magnet/ferromagnetic material or magnet/magnet interactions described herein.

[0060] Referring now to FIG. **8**, an embodiment of a method **800** for magnetically securing and releasing an optical connector to and from and an optical port is illustrated. As discussed below, the systems and methods of the present disclosure provide magnetic securing subsystems on an optical connector and optical port that are configured to magnetically interact to align and connect with each other to secure the optical connector to the optical port (while preventing their connection if the optical connector is misoriented with the optical port in some embodiments), along with a mechanical release device on the optical connector that is configured to mechanically disconnect the magnetic securing subsystems on the optical connector and the optical port to release the optical connector from the optical port. For example, the optical cable system of the present disclosure may include an optical cable and an optical connector that is included on the optical cable. A connector magnetic securing subsystem is included on the optical connector and is configured to magnetically connect to a port magnetic securing subsystem on an optical port to secure the optical connector to an optical port. A mechanical release device is coupled to the optical connector and is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port. As such, the securing and release of optical connectors and optical ports is enabled while eliminating the issues with conventional optical port/connector secure/release systems discussed above.

[0061] The method **800** begins at block **802** where an optical connector on an optical cable moves adjacent an optical port. With reference to FIG. **9**A, in an embodiment of block **802**, the optical connector **404** on the optical connector **404** included on the optical cable system **400** may be moved in a direction **900** and adjacent the optical port **504** on the optical transceiver system **500**, and one of skill in the art in possession of the present disclosure will appreciate how the optical connector **404** is illustrated in FIG. **9**A as being oriented with the optical port **504** in a manner that would provide for proper connection with the optical port **504**. With reference to FIG. **10**A, in another embodiment of block **802**, the optical connector **404** on the optical connector **404** included on the optical cable system **400** may be moved in a direction **1000** and adjacent the optical port **610** on the computing device **600**, and one of skill in the art in possession of the present disclosure will appreciate how the optical connector **404** is illustrated in FIG. **10**A as being oriented with the optical port **610** in a manner that would provide for proper connection with the optical port **610**.

[0062] With reference to FIG. **11**A, in another embodiment of block **802**, the optical connector **404** on the optical connector **404** included on the optical cable system **400** may be moved in a direction **1100** and adjacent the optical port **504** on the optical transceiver system **500**, and one of skill in the art in possession of the present disclosure will appreciate how the optical connector **404** is illustrated in FIG. **11**A as being misoriented with the optical port **504** in a manner that would not provide for proper connection with the optical port **504**. With reference to FIG. **12**A, in another embodiment of block **802**, the optical connector **404** on the optical connector **404** included on the optical cable system **400** may be moved in a direction **1200** and adjacent the optical port **610** on the computing device **600**, and one of skill in the art in possession of the present disclosure will appreciate how the optical connector **404** is illustrated in FIG. **12**A as being misoriented with the optical port **610** in a manner that would not provide for proper connection with the optical port **610**.

[0063] The method **800** then proceeds to decision block **804** where the method **800** proceeds depending on whether the optical connector is oriented with the optical port. In the embodiments illustrated and described below, the magnetic securing subsystems on the optical connector and optical port of the present disclosure are configured with a keying system (e.g., provided by the connector keying subsystem and port keying subsystem described above) to prevent securing of the optical connector to the optical port if they are misoriented, and thus the method **400** may proceed at decision block **804** depending on whether the optical connector and optical port or oriented or misoriented. However, one of skill in the art in possession of the present disclosure will appreciate how such functionality is optional in the optical port/connector magnetic securing/release system of the present disclosure, and thus the method **400** may proceed from block **802** directly to block **806** in embodiments in which the keying system is omitted.

[0064] If, at decision block **804**, the optical connector is oriented with the optical port, the method **800** proceeds to block **806** where a port magnetic securing subsystem on the optical port magnetically interacts with a connector magnetic securing subsystem on the optical connector to align the optical connector with the optical port. With reference to FIG. **9**B, in an embodiment of block **806** and following the movement of the optical connector **404** adjacent the optical port **504** when the optical connector **404** is oriented with the optical port **504** as described above with reference to FIG. **9**A, the connector magnetic securing subsystem on the optical connector **404** (e.g., provided by the magnets **408** and **410** in the illustrated embodiment) will magnetically interact with the port magnetic securing subsystem **504*a*** on the optical port **504** to produce a magnetic attraction force **902** that pulls the optical connector **404** towards the optical port **504**.

[0065] Furthermore, one of skill in the art in possession of the present disclosure will appreciate how the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504*a*** on the optical port **504** may be configured (e.g., via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding ferromagnetic material that provides the port magnetic securing subsystem **504*a*** on the optical port **504**, via magnets that provide the port magnetic securing subsystem **504*a*** on the optical port **504** and corresponding ferromagnetic material that provides the connector magnetic securing subsystem on the optical connector **404**, via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding magnets that provide the port magnetic securing subsystem **504*a*** on the optical port **504**, etc.) to cause the optical connector **404** to align with the optical port **504** (e.g., such that fiber ferrule(s) that are included in the optical conduit subsystem **406** and that are moveably coupled to the optical connector **404** via a spring align with fiber ferrule receiver(s) on the optical conduit subsystem **704** of the optical port **700** that provides the optical port **504**).

[0066] With reference to FIG. **10**B, in an embodiment of block **806** and following the movement of the optical connector **404** adjacent the optical port **610** when the optical connector **404** is oriented with the optical port **610** as described above with reference to FIG. **10**A, the connector magnetic securing subsystem on the optical connector **404** (e.g., provided by the magnets **408** and **410** in the illustrated embodiment) will magnetically interact with the port magnetic securing subsystem **610*a*** on the optical port **610** to produce a magnetic attraction force **1002** that pulls the optical connector **404** towards the optical port **610**. Similarly as described above, one of skill in the art in possession of the present disclosure will appreciate how the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610*a*** on the optical port **610** may be configured (e.g., via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding ferromagnetic material that provides the port magnetic securing subsystem **610*a*** on the optical port **610**, via magnets that provide the port magnetic securing subsystem **610*a*** on the optical port **610** and corresponding ferromagnetic material that provides the connector magnetic securing subsystem on the optical connector **404**, via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding magnets that provide the port magnetic securing subsystem **610*a*** on the optical port **610**, etc.) to cause the optical connector **404** to align with the optical port **610** (e.g., such that fiber ferrule(s) that are included in the optical conduit subsystem **406** and that are moveably coupled to the optical connector **404** via a spring align with fiber ferrule receiver(s) on the optical conduit subsystem **704** of the optical port **700** that provides the optical port **610**).

[0067] If, at decision block **804**, the optical connector is misoriented with the optical port, the method **800** proceeds to block **808** where the port magnetic securing subsystem on the optical port magnetically prevents the connector magnetic securing subsystem on the optical connector from securing the optical connector to the optical port. With reference to FIG. **11**B, in an embodiment of block **806** and following the movement of the optical connector **404** adjacent the optical port **504** when the optical connector **404** is misoriented with the optical port **504** as described above with reference to FIG. **11**A, the connector magnetic securing subsystem on the optical connector **404** (e.g., provided by the magnets **408** and **410** in the illustrated embodiment) will magnetically interact with the port magnetic securing subsystem **504*a*** on the optical port **504** to produce a magnetic repulsion force **1102** that pushes the optical connector **404** away the optical port **504**.

[0068] As discussed above, the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504*a*** on the optical port **504** may be configured (e.g., via the magnets **306*a*** and **308*a*** that provide the connector magnetic securing subsystem on the optical connector **300** that provides the optical connector **404**, and the magnets **706*a*** and **708*a*** that provide the port magnetic securing subsystem on the optical port **700** that provides the optical port **504**, etc.) to prevent the optical connector **404** from connecting and securing with the optical port **504** when the optical connector **404** is misoriented with the optical port **504**. One of skill in the art in possession of the present disclosure will appreciate how the movement of the magnets **306***a* and **706***a* that each have their north pole end exposed, as well as the movement of the magnets **306***b* and **706***b* that each have their south pole end exposed, adjacent each other will produce the magnetic repulsion force that will prevent engagement of the optical connector **404/300** and the optical port **504/700** (i.e., while also appreciating how the movement of the magnets **306***a* and **706***b*, as well as the movement of the magnets **306***b* and **706***a*, adjacent each other will produce the magnetic attraction force discussed above that will align and engage of the optical connector **404/300** and the optical port **504/700**).

[0069] With reference to FIG. **12**B, in an embodiment of block **806** and following the movement of the optical connector **404** adjacent the optical port **610** when the optical connector **404** is misoriented with the optical port **610** as described above with reference to FIG. **12**A, the connector magnetic securing subsystem on the optical connector **404** (e.g., provided by the magnets **408** and **410** in the illustrated embodiment) will magnetically interact with the port magnetic securing subsystem **610***a* on the optical port **610** to produce a magnetic repulsion force **1202** that pushes the optical connector **404** away the optical port **610**. As discussed above, the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610***a* on the optical port **610** may be configured (e.g., via the magnets **306***a* and **308***a* that provide the connector magnetic securing subsystem on the optical connector **300** that provides the optical connector **404**, and the magnets **706***a* and **708***a* that provide the port magnetic securing subsystem on the optical port **700** that provides the optical port **610**, etc.) to prevent the optical connector **404** from connecting and securing with the optical port **610** when the optical connector **404** is misoriented with the optical port **610**. One of skill in the art in possession of the present disclosure will appreciate how the movement of the magnets **306***a* and **706***a* that each have their north pole end exposed, as well as the movement of the magnets **306***b* and **706***b* that each have their south pole end exposed, adjacent each other will produce the magnetic repulsion force that will prevent engagement of the optical connector **404/300** and the optical port **610/700** (i.e., while also appreciating how the movement of the magnets **306***a* and **706***b*, as well as the movement of the magnets **306***b* and **706***a*, adjacent each other will produce the magnetic attraction force discussed above that will align and engage of the optical connector **404/300** and the optical port **610/700**).

[0070] Following block **806**, the method **800** proceeds to block **810** where the port magnetic securing subsystem on the optical port magnetically connects to the connector magnetic securing subsystem on the optical connector to secure the optical connector to the optical port. With reference to FIG. **9**C, in an embodiment of block **810** and in response to moving the optical connector **404** adjacent the optical port **504** such that the magnetic attraction force **902** discussed above with reference to FIG. **9**B is produced, the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504***a* on the optical port **504** may magnetically connect (e.g., via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding ferromagnetic material that provides the port magnetic securing subsystem **504***a* on the optical port **504**, via magnets that provide the port magnetic securing subsystem **504***a* on the optical port **504** and corresponding ferromagnetic material that provides the connector magnetic securing subsystem on the optical connector **404**, via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding magnets that provide the port magnetic securing subsystem **504***a* on the optical port **504**, etc.) to secure the optical connector **404** to the optical port **504** (e.g., such that fiber ferrule(s) that are included in the optical conduit subsystem **406** and that are moveably coupled to the optical connector **404** via a spring align with fiber ferrule receiver(s) on the optical conduit subsystem **704** of the optical port **700** that provides the optical port **504**).

[0071] With reference to FIG. **10**C, in another embodiment of block **810** and in response to moving the optical connector **404** adjacent the optical port **610** such that the magnetic attraction force **1002** discussed above with reference to FIG. **10**B is produced, the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610***a* on the optical port **610** may magnetically connect (e.g., via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding ferromagnetic material that provides the port magnetic securing subsystem **610***a* on the optical port **610**, via magnets that provide the port magnetic securing subsystem **610***a* on the optical port **610** and corresponding ferromagnetic material that provides the connector magnetic securing subsystem on the optical connector **404**, via the magnets **408** and **410** that provide the connector magnetic securing subsystem on the optical connector **404** and corresponding magnets that provide the port magnetic securing subsystem **610***a* on the optical port **610**, etc.) to secure the optical connector **404** to the optical port **610** (e.g., such that fiber ferrule(s) that are included in the optical conduit subsystem **406** and that are moveably coupled to the optical connector **404** via a spring align with fiber ferrule receiver(s) on the optical conduit subsystem **704** of the optical port **700** that provides the optical port **610**).

[0072] The method **800** then proceeds to decision block **812** where the method **800** proceeds depending on whether the optical cable is to-be disconnected from the optical port. As will be appreciated by one of skill in the art in possession of the present disclosure, following the connection of an optical cable to an optical port, a variety of reasons may arise in which that optical cable must be disconnected from that optical port, and thus the method **800** may proceed depending on whether the optical cable **400** must be disconnected from the optical port **504** or **610** and, thus, whether the mechanical release device of the present disclosure must be utilized. If, at decision block **812**, the optical cable is not to-be disconnected from the optical port, the method **800** returns to block **810**. As such, the method **800** may loop such that the port magnetic securing subsystem on the optical port remains magnetically connected to the connector magnetic securing subsystem on the optical connector to secure the optical connector to the optical port as long as the optical cable is not to-be disconnected from the optical port.

[0073] If, at decision block **812**, optical cable is to-be disconnected from the optical port, the method **800** proceeds to block **814** where a mechanical release device coupled to the optical connector is actuated to mechanically disconnect the connector magnetic securing subsystem on the optical connector from the port magnetic securing subsystem on the optical port and release the optical connector from the optical port. With reference to FIGS. **13**A, **13**B, and **13**C, in an embodiment of block **814** and with the mechanical release device on the optical connector **404** in the securing orientation A while the optical connector **440** is secured to the optical port **504**, a user of the optical cable system **400** may apply the force F discussed above with reference to FIG. **4**C to the actuator element **412** that is sufficient to compress the spring device **412***a*, which as described above causes the arm members **414** and **416** to rotate in a direction R about their respective hinges **414***a* and **416***a* such that their respective port engagement portions **414***b* and **416***b* move away from the front surface **404***c* of the optical connector **404** to position the mechanical release device on the optical connector **404** in the release orientation B, and one of skill in the art in possession of the present disclosure will appreciate how such movement of the port engagement portions **414***b* and **416***b* on the mechanical release device will cause those port engagement portions **414***b* and **416***b* to engage a surface of the optical port **504** to provide a release force that is directed away from the optical port **504** from the point of view of the optical connector **404**.

[0074] Furthermore, one of skill in the art in possession of the present disclosure will appreciate how the mechanical advantage provided by the dimensions of the mechanical release device may be configured to produce a release force that is sufficient to overcome the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504***a* on the optical port **504** (e.g., a relatively significant magnetic attraction force that may be produced via the engagement of corresponding neodymium magnets used in at least some of the embodiments described above.) In response to actuating the mechanical release device and providing the release force that overcomes the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504***a* on the optical port **504**, the user may move the optical connector **404** in a direction **1300** and away from the optical port **504** (while continuing to provide the force F on the actuator element **412** to ensure the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **504***a* on the optical port **504** does not cause their reconnection). As illustrated in FIG. **13**C, the user may then cease providing the force F on the actuator element **412**, with the spring element **412***a* resiliently biasing the mechanical release device on the optical connector **404** from the release orientation B back into the securing orientation A.

[0075] With reference to FIGS. **14**A, **14**B, and **14**C, in an embodiment of block **814** and with the mechanical release device on the optical connector **404** in the securing orientation A while the optical connector **440** is secured to the optical port **610**, a user of the optical cable system **400** may apply the force F discussed above with reference to FIG. **4**C to the actuator element **412** that is sufficient to compress the spring device **412***a*, which as described above causes the arm members **414** and **416** to rotate in a direction R about their respective hinges **414***a* and **416***a* such that their respective port engagement portions **414***b* and **416***b* move away from the front surface **404***c* of the optical connector **404** to position the mechanical release device on the optical connector **404** in the release orientation B, and one of skill in the art in possession of the present disclosure will appreciate how such movement of the port engagement portions **414***b* and **416***b* on the mechanical release device will cause those port engagement portions **414***b* and **416***b* to engage a surface of the optical port **610** and provide a release force that is directed away from the optical port **610** from the point of view of the optical connector **404**.

[0076] Similarly as described above, one of skill in the art in possession of the present disclosure will appreciate how the mechanical advantage provided by the dimensions of the mechanical release device may be configured to produce the release force that is sufficient to overcome the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610***a* on the optical port **610** (e.g., a relatively significant magnetic attraction force that may be produced via the engagement of corresponding neodymium magnets used in at least some of the embodiments described above.) In response to actuating the mechanical release device and providing the release force that overcomes the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610***a* on the optical port **610***a*, the user may move the optical connector **404** in a direction **1400** and away from the optical port **610** (while continuing to provide the force F on the actuator element **412** to ensure the magnetic attraction force between the connector magnetic securing subsystem on the optical connector **404** and the port magnetic securing subsystem **610***a* on the optical port **610** does not cause their reconnection). As illustrated in FIG. **14**C, the user may then cease providing the force F on the actuator element **412**, with the spring element **412***a* resiliently biasing the mechanical release device on the optical connector **404** from the release orientation B back into the securing orientation A.

[0077] As will be appreciated by one of skill in the art in possession of the present disclosure, the mechanical release device provided on the optical connectors of the present disclosure may be particularly useful in embodiments that utilize neodymium magnets in the magnetic securing subsystem(s) on the optical connector and/or the optical port, as the small size of magnets required for the optical port/connector connection/securing applications of the present disclosure combined with the high strength of neodymium magnets presents issues with those magnets breaking off from the component to which they are attached when an attempt is made to overcome the attractive magnetic force which they produce. For example, the magnetic attraction force produced by small neodymium magnets may be greater than the magnet/component attachment techniques used to attach those neodymium magnets to the optical connector and/or optical port, and a user pulling on the optical cable that includes the optical connector of the present disclosure may simply rip the magnet off the optical connector or optical port, or damage another part of the optical cable system before producing a force sufficient to overcome the attractive magnetic force produced by the neodymium magnets(s), and one of skill in the art in possession of the present disclosure will appreciate how the mechanical release device operates to prevent such optical cable system component damage.

[0078] Thus, systems and methods have been described that provide magnetic securing subsystems on an optical connector and optical port that are configured to magnetically interact to align and connect with each other to secure the optical connector to the optical port (while preventing their connection if the optical connector is misoriented with the optical port in some embodiments), along with a mechanical release device on the optical connector that is configured to mechanically disconnect the magnetic securing subsystems on the optical connector and the optical port to release the optical connector from the optical port. For example, the optical cable system of the present disclosure may include an optical cable and an optical connector that is included on the optical cable. A connector magnetic securing subsystem is included on the optical connector and is configured to magnetically connect to a port magnetic securing subsystem on an optical port to secure the optical connector to an optical port. A mechanical release device is coupled to the optical connector and is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port. As such, the securing and release of optical connectors and optical ports is enabled while eliminating the issues with conventional optical port/connector secure/release systems discussed above.

[0079] Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

What is claimed is:

1. An optical cable system, comprising:

an optical cable;

an optical connector that is included on the optical cable;

a connector magnetic securing subsystem that is included on the optical connector and that is configured to magnetically connect to a port magnetic securing subsystem on an optical port to secure the optical connector to an optical port; and

a mechanical release device that is coupled to the optical connector and that is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port.

2. The system of claim **1**, wherein the connector magnetic securing subsystem includes a plurality of neodymium magnets.

3. The system of claim **1**, wherein the optical connector includes an optical conduit subsystem, and wherein the mechanical release device is located adjacent the optical conduit subsystem when in a securing orientation while the optical connector is secured to the optical port, and includes a moveable connection to the optical connector that configures the mechanical release device to move relative to the optical connector and into a release orientation in which the mechanical release device extends from the optical connector and into engagement with the optical port to release the optical connector from the optical port.

4. The system of claim **1**, wherein the connector magnetic securing subsystem includes a plurality of connector magnets that are configured to magnetically interact with the port magnetic securing subsystem to align the optical connector with the optical port.

5. The system of claim **4**, wherein the port magnetic securing subsystem includes respective ferromagnetic elements for each of the plurality of connector magnets.

6. The system of claim **1**, wherein the connector magnetic subsystem includes a plurality of connector magnets that are configured to magnetically interact with corresponding port magnets included in the port magnetic securing subsystem to align and connect the optical connector with the optical port when the optical connector is oriented with the optical port, and that are configured to magnetically interact with the corresponding port magnets included in the port magnetic securing subsystem to prevent connection of the optical connector with the optical port when the optical connector is misoriented with the optical port.

7. An Information Handling System (IHS), comprising:

a chassis;

a processing system that is housed in the chassis;

an optical port that is included on the chassis and that is coupled to the processing system;

a port magnetic securing subsystem that is included on the optical port;

an optical cable;

an optical connector that is included on the optical cable;

a connector magnetic securing subsystem that is included on the optical connector and that is magnetically connected to the port magnetic securing subsystem to secure the optical connector to the optical port; and

a mechanical release device that is coupled to the optical connector and that is configured to mechanically disconnect the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port.

8. The IHS of claim **7**, wherein the connector magnetic securing subsystem includes a plurality of neodymium magnets.

9. The IHS of claim **7**, wherein the optical connector includes an optical conduit subsystem, and wherein the mechanical release device is located adjacent the optical conduit subsystem when in a securing orientation while the optical connector is secured to the optical port, and includes a moveable connection to the optical connector that configures the mechanical release device to move relative to the optical connector and into a release orientation in which the mechanical release device extends from the optical connector and into engagement with the optical port to release the optical connector from the optical port.

10. The IHS of claim **7**, wherein the connector magnetic securing subsystem includes a plurality of connector magnets that are configured to magnetically interact with the port magnetic securing subsystem to align the optical connector with the optical port when the optical connector is moved adjacent the optical port.

11. The IHS of claim **10**, wherein the port magnetic securing subsystem includes respective ferromagnetic elements for each of the plurality of connector magnets.

12. The IHS of claim **11**, wherein a portion of the chassis adjacent port magnetic securing subsystem is provided by a non-ferromagnetic material.

13. The IHS of claim **7**, wherein the connector magnetic subsystem includes a plurality of connector magnets that are configured to magnetically interact with corresponding port magnets included in the port magnetic securing subsystem to align and connect the optical connector with the optical port when the optical connector is moved adjacent the optical port while oriented with the optical port, and that are configured to magnetically interact with the corresponding port magnets included in the port magnetic securing subsystem to prevent connection of the optical connector with the optical port when the optical connector is moved adjacent the optical port while misoriented with the optical port.

14. A method for magnetically securing and releasing an optical connector to and from an optical port, comprising:

moving, by an optical connector that is included on an optical cable, adjacent an optical port;

magnetically connecting, by a connector magnetic securing subsystem that is included on the optical connector, to a port magnetic securing subsystem on the optical port to secure the optical connector to the optical port; and

mechanically disconnecting, by a mechanical release device that is coupled to the optical connector, the connector magnetic securing subsystem from the port magnetic securing subsystem to release the optical connector from the optical port.

15. The method of claim **14**, wherein the connector magnetic securing subsystem includes a plurality of neodymium magnets.

16. The method of claim **14**, further comprising:

moving, by the mechanical release device from a securing orientation that is adjacent an optical conduit subsystem on the optical connector while the optical connector is secured to the optical port, relative to the optical connector and into a release orientation in which the mechanical release device extends from the optical connector and into engagement with the optical port to release the optical connector from the optical port.

17. The method of claim **14**, further comprising:

magnetically interacting, by a plurality of connector magnets included on the connector magnetic securing subsystem, with the port magnetic securing subsystem to align the optical connector with the optical port when the optical connector is moved adjacent the optical port.

18. The method of claim **17**, wherein the port magnetic securing subsystem includes respective ferromagnetic elements for each of the plurality of connector magnets.

19. The method of claim **18**, wherein the optical port is included on a chassis, and wherein a portion of the chassis adjacent port magnetic securing subsystem is provided by a non-ferromagnetic material.

20. The method of claim **14**, further comprising:

magnetically preventing, by a plurality of connector magnets included in the connector magnetic securing subsystem via magnetic interaction with corresponding port magnets included in the port magnetic securing subsystem, connection of the optical connector with the optical port when the optical connector is moved adjacent the optical port while misoriented with the optical port; and

magnetically connecting, by the plurality of connector magnets via magnetic interaction with the corresponding port magnets, the optical connector with the optical port when the optical connector is moved adjacent the optical port while oriented with the optical port.

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